

REFERENCES

No	Document	Description
1	5.2.2.1 - Site Specific Information_Rev 1	Owner Growth Project - Process Plant - Site Specific Information
2	4.7.1 - Consultation Report 7555 - Owner PDC Data Summary Rev 0, March 2022	Table 2-5, "Back Calculated Ore Parameters", 2014 - 2021 value.
4	4.6.1 - Owner Growth Mass Balance_Nominal Case Report_Rev C	Mass Balance - SysCAD Report Nominal Case. 22/03/2022
5	Owner Comminution Plant Data_B.xlsx"	Benchmarking from FIM pebble crushing circuit.
7	ECI Tender from Concep	Q12375 - EPC Gravity Circuit Rev1
8	5.1.12f - BOD Alignment Template - Flotation Tail PLT, FT CIL, Elution & EW	Basis of Design for Flotation Tailings CIL, Elution & EW circuit. 06/04/2022
9	4.7.3 - Consultation Report 7555 Owner PDC PSD Addendum Rev 1	Owner - Process Design Criteria Summary - PSD Addendum Report No. 7555 Rev 1 March 2022
10	5.1.12g - BOD Alignment Template - Reagents and Services	5.1.12g - BOD Alignment Template - Reagents and Services - Rev C Owner confirmation that regrind mill ball size will be 25 mm (cell F15).
11	EPC Calculation CAL-PR-002	Flotation Tailings CIL Simulation @ 8 and 10 g/L Carbon Concentration
14	Owner Growth Project ECI Outcome Letter - EPC	Owner Growth Project - Process Plant - Formal Agreement for Engineering, Procurement and Construction: Early Contractor Involvement (Value Engineering Phase). 03/08/2022
15	5.1.12b - BOD Alignment Template - 13 Grinding - Rev C	5.1.12b - BOD Alignment Template -13 Grinding - Rev C - EPC (220609)_NSR Review
16	Owner Process Design Criteria - VEP Phase (Stage 1)_I	Owner - Growth Project - Process Design Criteria Rev I. 20/09/2022
18	Cleaner Option 2- Owner Case 2 Scav Con to Cleaner	Glencore Technology - Jameson Cell Mass Balance - Cleaner
19	Cleaner Scavenger - Owner Case 2 Scav to Cleaner	Glencore Technology - Jameson Cell Mass Balance - Cleaner Scavenger
23	Owner GP Process Design Criteria - Reliance Information S1 Rev K, 25/01/2023	Principal supplied reliance information.
24	VEP Outcome Letters - EPC 09/02/23 and 020523	Principal supplied Scope of Works letter for Stage 1 and 2
25	Consultation Report 7653 - Owner Value Engineering Update	Grinding circuit modelling for Owner by Orway Mineral Consultants, 160223
26	Process Design Workshop Meeting Minutes 170223, MOM-PM-SFP002	Workshop between the Client and EPC
27	Glencore Jameson Cell Sizings and Quote	Simulations from Vendor, 22/02/24 Owner Expansion Project - Jameson Cell Quote OTJ-2220 7532.3
28	Weir Cyclone Simulations and Datasheet - Primary & Gravity Cyclones	Various simulations for the different process cases, and datasheet: 2000-F00-DSH-ME-11005A_B Primary Cyclone
31	Trash Screen Spray Water Information from Vendor	Emails with Vendor, 16/08/23
32	Flotation Mass Balance Scenarios	Mass Balances for Process Margins Rev H and New Mass Balance Cases Rev G
33	Cadia Gravity Electrowinning Cell	Emails with Vendor, 14/03/23. Ag and Cu Assays from site, provided by Owner (23/03/23)
35	Discussions with Client and TQ regarding Cyanide Destruction	Emails with Client, 230323 and TQ: 2000-F72-TQY-ME-10094_0_TQ-R
36	Vendor Information from Transmin for New Flocculant Mixing Plant	Transmin Proposal Execution Phase rev 3, P33-904-06
37	Direction from EPC Mechanical Engineers re: COS	Email, 280323
38	Client Information regarding Grinding Media	Emails with Client, 070323
39	Vendor Information from Concep for the Acacia CS8000 ILR	Emails with Client, 210322
40	EPC Calculation	"SAG_Ball_Screen_Trommel_Calc_Rev_A.xlsx"
41	Client Direction, Transfer of FLS Equipment Technical Information	Emails from client, various dates (May 2023)
42	Primary Crusher Datasheet	Scope of Work - Primary Crusher, Appendix A, doc 2000-F12-SOW-ME-20001 rev 1
43	SoW for Separable Portions Stage 1&2	Scope of Works between Owner and EPC 2000-F00-SOW-CO-20001 & 2000-F00-SOW-CO-20002_0
44	Outstanding Process Inputs from Owner	Outstanding Process Inputs 210623 (version 1) Additional (003) (002), and 2000 F00-TQY-PR-10051.0.TQ-R.0.01
47	Oxygen Consumption Information	Emails from client, 8/5/23
48	Cadia CIL4 Electrowinning Cells	Vendor Quotation 27/03/2024
49	Owner email Owner GP - Float Cell Motors", dated 11 May 2023	FLSmidth flotation blower details.
50	Calculation for LPG Consumption and TPE ECI Tender	CAL-PR-005_A LPG Consumption TPE5000 ELUTION HEATER PACKAGE - ME16 TENDER - COMPLETE 20/05/22
51	FLSmidth - Primary Crusher Application Model	FLS_13x19 TSUV_Calc_FIM_Run 15_13x19 TSUV_ 6.5 OSS_3716 mtph_70, AVG results
52	Vendor Information Mercury Abatement System (CIL4 Carbon Regen Kiln) and Client Direction Variation 25; Vendor information for Lochhead CIL4 Carbon Regen Kiln; Regen Kiln Off Gas Calculations	Information from the Mercury Abatement Vendor, Jan 2024 P3-EPC-COR-161 Variation Direction No.025 Carbon regeneration kiln off-gas mercury abatement system Lochhead Carbon Reactivation Kiln (ME-17) Quotation No. 5211 (August 26, 2022) Kiln Off Gas Calc
53	EPC email Apron Feeder Datasheet Internal Review", dated 30 August 2023	Agreement with Owner to change contract SOW design factor from 120% to 110% of nominal tonnage rate.
54	Weir UFG Cyclone Cluster Modelling	UFG_CY_Select_Design_RevC UFG_CY_Select_Nominal_RevC
55	Takraf Tender Rev 0 Datasheet	2000-F00-DSH-ME-11018_B Trash Screens Completed
56	Owner GP Process Design Criteria - Reliance Information S1 Rev L, 25/01/2023	Principal supplied reliance information.
57	Owner Growth Project SysCAD Model (Historical)	SysCAD Model files and Reports (2022.08.31)-01.spf
58	NTF Deslime & UFG Cyclone PDC Inputs	Word document 260723
59	Joest Vibrating Screens Offer	JOEST_M22-11102 Rev D_JOEST FIRM Offer 2000-F00-DSH-ME-11011_C Vibrating Screens
60	Flotation Launder Wash Water Technical Query	Technical Query (2000-F16-TQY-ME-10007) FLS Launder Water Requirements Rev.0 31/07/2023
62	Technical Query - Design Bulk Density Range for Apron Feeders And Conveyors	2000-F00-TQY-ME-10034

REFERENCES

No	Document	Description
63	Flotation Area - Design P80 Sizes	Email Correspondence. RE - Flotation D80 Mech Requests. 250823
64	Reagent Delivery Formats, 2000-F72-TQY-ME-20001_2	Reagent delivery information, provided by Owner
65	IsaMill Quote and Modelling by Glencore	IsaMill Quote OMT232007747 Owner - Owner EPC - IsaMill Sizing - Design -19102023 - New data supplied water temp 25 Owner - Owner EPC - IsaMill Sizing - Nominal -19102023 - New data supplied water temp 25 Email: "FW_ ME039 IsaMill Latest MB - Clarifications 64 - UFG Mill Media Hopper Live Volume"
66	Regrind Removal TQ	2000-F00-TQY-GE-10080_0 - RESPONSE.pdf
67	Client Comments Stage 1 Rev K PDC and Rev L SysCAD	2000-F00-DCR-PR-10001.K.ICR-R.K.02
68	Raw Water TQ 10044	2000-F24-TQY-PR-10044.0.TQ-R.0.0164
69	SysCAD & PDC Client Comments TQ 10107 Response	2000-F00-TQY-PR-10107.0.TQ-R.0.02
70	Gravity Variation (VAR-017) and Communication from the Client	P3-EPC-COR-085 Variation Direction No.017 Gravity Circuit Modifications Emails from Client/Outcome of meetings
71	Emails from Concep (Knelson & ILR) in relation to Gravity Variation (VAR-017)	Email dated 30/10/23, 'RE: Knelson Concentrators (KC-QS70) and CS10000'
72	Gravity Variation TQ 10115 Response from Client	2000-F14-TQY-PR-10115.0.TQ-R.0.01 2000-F14-TQY-PR-10115_1.TQ-R
73	Quotation and emails from Cadia Systems (EW cells) in relation to Gravity Variation (VAR-017)	Q231115-PR1_EPC_EWC 151123 issue RE_ Owner - Gravity Pregnant Solution Electrowinning - various emails
74	Variation 018 UFG Mill 2 Trash Screen	P3-EPC-COR-104 Variation Direction No.018 UFG Mill 2 Trash Screen
75	Clarification Emails for Flotation Reagent Calculations	Emails with Client 21/11/23: 'RE: Owner GP - PAX Usage/Flow Consumption/Storage Tank RT'
76	Dump Pocket TQ from Owner	2000-F12-TQY-GE-10127_0.TQ-R
77	Wet Static Screens Datasheet (ME40) and Spray Water Information	2000-F00-DSH-ME-11040_1 and emails dated 23/01/24
78	Owner Owner's Manual CAT793F Owner	Owner's Manual CAT793F Owner
79	Water Services PFD Client Review	2000-F00-PFD-PR-10046.D.IFD-R.D.02
80	Area 30 M1 Design Review Report	2000-F30-REP-GE-10101_A - M1 Design Review Report
81	TPE CIL4 Elution Heater Vendor Package	TPE5000 ELUTION HEATER PACKAGE_DES.pdf rev A 20/05/22
82	Client Information LPG Storage Vessel	Email: RE LPG Tank, 22/05/2023
83	TQ147 Reagent Tank Volumes, Var 26 Cyanide Tanks and Var 192 CuSO4 Tanks	2000-F72-TQY-ME-10147_0.TQ-R P3-EPC-COR-162 Variation Direction No.026 Cyanide Tank Storage Capacity P3-EPC-COR-192 Variation Direction No.030 Copper Sulphate Storage
84	Joest Tender Documents	JOEST_M22-11102 Rev H_JOEST FIRM Offer, 06/05/24 RE Growth Project - Stage 2 Vibrating Screen Changes
85	Gekko Carbon Scout ECI Tender Documents	6454-EPC-Carbon Scout-220426, 22/04/22
86	ME65 Carbon Transfer Pump Vendor Information	Weir Budget Quote WMFN20220527
87	ME07 Intertank Screens Vendor Information	P28337_Northern Star_EPC Group Growth (Aus) REV 2
88	TQ206 Request for LPG & RTO Information	2000-F00-TQY-PR-10206.0.TQ-R.0.0270
89	2000-F00-CAL-ME-F23TK404 Final Tails Tank Live Vol & OF Calc.xlsx	2000-F00-CAL-ME-F23TK404 Final Tails Tank Live Vol & OF Calc.xlsx
90	Client Comments Stage 1 Rev 0 PDC	2000-F00-DCR-PR-10001.0.IFD-R.0.02
91	Stage 2 Oxygen Consumption Calculations	CAL-PR-003 S2 Oxygen Consumption Calculations
92	SAG & BM Trommel Sizes	2000-ME001.2-B02-50041_C SAG Mill Trommel Sizing Report 2000-ME001.2-M03-50001_B SAG Mill Trommel Assembly 2000-ME001.3-B02-50020_B Ball Mill Trommel Sizing Report 2000-ME001.3-M03-50001_B Ball Mill Trommel Assembly
100	5.1.1 - ECS Process Design Criteria	EngineerProcess Design Criteria - Owner Growth Project ECS. 25/01/2022 J6027-000-DCR-PR-001 Rev D
101	5.1.2 - ECS Mass Balance	EngineerMass Balance - Owner Growth Project ECS. 25/01/2022 J6027-000-CAL-PR-001 Rev D
102	4.5.1 - Process Design Criteria Rev F	Owner - Growth Project - Process Design Criteria Rev F. 16/03/2022
103	4.5.1 - Owner Process Design Criteria - VEP Phase _H, 19/08/2022	Principal supplied reliance information.
104	Email "Carbon Safety Screen" dated 16/02/2024	Confirmation from Owner that currently proposed screen sizes for Carbon Safety Screens is adequate
105	2000-F30-TQY-PR-15001.0.TQ-R.0.02	Client Response to Technical Query - CIL2/3 - Carbon Transfer Rate Between Adsorption Tanks
106	2000-F00-DCR-PR-10002.C.ICR.-R.C.02	Client comments to Process Design Criteria - Stage 2
107	2000-F72-TQY-ME-10260_0-R	Clients Response to Technical Query - Milk of Lime Tank Volume & Residence Time
108	2000-F21-TQY-PR-15014.0.TQ-R.0.02	Clients Response to Technical Query - CIL2/3 & CIL4 Target Process Data
109	2000-F00-TQY-PR-15020.0.TQ-R.0.02	Clients Response to Technical Query - CIL4 Carbon Advance
110	2000-F22-TQY-PR-15018_1_RSP	Clients Response to Technical Query - HCI Dilution Tank
111	Hydrogen Peroxide Technical Data Sheet	Solvay Hydrogen Peroxide Technical Datasheet

REFERENCES

No	Document	Description
112	PAX, CuSO4 & Lead Nitrate Delivery Sizes	Email from IXOM (vendor), dated 28/03/2024
113	Concentrate Thickener Vendor GA	2000-ME091-B02-10001_C
114	Client Comments Stage 2 Rev 0 PDC	20000-F00-DCR-PR-10002_0_AAN
115	2000-F30-TQY-PR-15029_0_TQ-R	Clients Response to Technical Query - Concentrate Pre-Leach Thickener Underflow
116	2000-F72-TQY-PR-15033_0_TQ-R	Clients Response to Technical Query - Lime System Process Design
117	QUO-07322-P7S5G0	Derrick quotation for Derrick Linear Motion Screening Machine 15/08/2024
118	BHT Lime Staking Package	2000-F00-DSH-ME-12013_0 2000-MEP-ME013-REG_A 706-PC-01 706-PC-02 FW_Lime Mill Media Consumption.msg
119	2000-F30-TQY-PR-15097_1	CIL3 Carbon Movement TQ

AREA	PARAMETER	UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
				1	2	3	4					
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
1.0	PRODUCTION SCHEDULE											
1.1	Throughput											
	Annual Throughput	t/a	10,500,000	25,500,000	23,997,254	25,500,000	28,350,000	1	Owner	16, 24, 26 & 32		A
	Annual Operating Days	d/a	365	365	365	365	365	1	Owner	16		A
1.2	Operating Schedules											
	Crushing											
	Plant Availability	%	65.0	65.0	65.0	65.0	65.0	1	Owner	1, 30		A
	Annual Operating Hours	h/a	5,694	5,694	5,694	5,694	5,694	6	Calculated			
	Hourly Throughput - Existing Circuit	th	1,844	1,844	1,844	1,844	1,844	6	Calculated			
	Hourly Throughput - New Circuit	th		2,634	2,370	2,634	3,135	6	Calculated			
	Total Hourly Throughput	th	1,844	4,478	4,214	4,478	4,979	6	Calculated			
1.3	Concentrator											
	Plant Availability	%	93.3	93.4	93.4	93.4	93.4	1	Owner	16, 23		A
	Annual Operating Hours - Existing SAG Circuit		8,000	-	-	-	-	6	Calculated			
	Annual Operating Hours - New SAG/Ball Mill Circuit	h/a	-	8,182	8,182	8,182	8,182	6	Calculated			
	Hourly Throughput - Existing SAG Circuit	th	1,313	1,313	1,313	1,313	1,313	6	Calculated			
	Hourly Throughput - New SAG/Ball Mill Circuit	th		1,833	1,650	1,833	2,182	6	Calculated			
	Total Hourly Throughput	th	1,313	3,146	2,962	3,146	3,494	6	Calculated			
	UFG Circuit											
	Plant Availability	%	93.3	93.4	93.4	93.4	93.4	1	Owner	16, 23		A
	Annual Operating Hours	h/a	8,000	8,182	8,182	8,182	8,182	6	Calculated			
2.0	MATERIAL CHARACTERISTICS											
2.1	General Ore Properties											
	Moisture Content	%w/w			1.3			1	Owner	16		A
	Ore Specific Gravity	Nominal			2.90			1	Owner	16		A
		Range			2.70 - 3.10			1	Owner	23		B
	Ore Bulk Density	Nominal			1.90			1	Owner	16		A
		Range			1.70 - 2.10			1	Owner	56		B
		Design Range									To be used for all conveyors & apron feeders in the crushing and milling circuits	B
	Angle of Repose	degrees			1.60 - 1.90			1	Owner	62		A
	Drawdown Angle	degrees			36			1	Owner	16		A
					60			1	Owner	16		A
2.2	Ore Comminution Parameters											
	Crushing Work Index											
	Testwork Average	kWh/t			17.8			1	Owner	16		A
	Testwork Range	kWh/t			16 - 30			1	Owner	16		A
	Testwork 80th Percentile	kWh/t			18.8			1	Owner	2, 23	2014 - 2021 value (as of rev E)	A
	Abrasion Index											
	Testwork Average				0.145			1	Owner	16		A
	Testwork Range				0.020 - 0.680			1	Owner	16		A
	Testwork 80th Percentile				0.192			1	Owner	16		A
	Ball Mill Work Index	Range	kWh/t		11.5 - 15			1	Owner	41		B
	A*b	Design	kWh/t		12.5			1	Owner	2	2014 - 2021 value	A
		Range			38 - 48			1	Owner	41		B
		Design			43.8			1	Owner	2	2014 - 2021 value	A
2.3	Pebble Comminution Parameters											
	Crushing Work Index	Design	kWh/t		18.8			1	Owner	16		A
		Maximum	kWh/t		23.0			1	Owner	23		A
	Bulk Density	Minimum	tm ³		1.60			1	Owner	16, 62		A
		Maximum	tm ³		1.90			1	Owner	41, 62		B
2.4	Non Ore Grades											
	Feed Grade - S	% S	1.02	1.02		1.40	1.02	1	Owner	23	Achievable grade back calculated by EPC (Source 32)	A
	Feed Grade - Au	Au (g/t)	1.58	1.58		1.70	1.58	1	Owner	23	Achievable grade back calculated by EPC (Source 32)	A
	Final Tail Grade - S	%	0.18	1.01	1.39		1.01	1	Owner	23	Blue data is from Owner Ref 23	A
	Final Tail Grade - Au	Au (g/t)	0.23	0.23	0.25		0.23	1	Owner	23	Blue data is from Owner Ref 23	A
2.5	Design Ore Grades											
	Feed Grade - S	%S	1.49		1.49			1	Owner	23		A
	Feed Grade - Au	Au (g/t)	1.81		1.81			1	Owner	23		A
	Final Tail Grade - S	%	0.29		1.48			1	Owner	23	Blue data is from Owner Ref 23	A
	Final Tail Grade - Au	Au (g/t)	0.38		0.26			1	Owner	23	Blue data is from Owner Ref 23	A
2.6	Recovery											
	Total Gold Recovery	Nominal	%	84	84	84	84	6	Calculated			
	Gravity Gold Recovery	Nominal	%	14.00	14.00	14.00	14.00	1	Owner	23	Relative to feed grade	A
	Gravity Gold Recovery	Maximum	%	20.00	20.00	20.00	20.00	1	Owner	23	Relative to feed grade	A
	Flotation Gold Recovery	Nominal	%	71.80	71.80	71.80	71.80	1	Owner	23	Relative to feed grade	A
	Flotation Gold Recovery	Maximum	%	75.00	75.00	75.00	75.00	1	Owner	23	Relative to feed grade	A
	Flotation Sulphur Recovery	Nominal	%	85.10	85.10	85.10	85.10	1	Owner	23	Relative to feed grade	A
	Flotation Sulphur Recovery	Maximum	%	90.00	90.00	90.00	90.00	1	Owner	23	Relative to feed grade	A
	Flotation Concentrate Leach Gold Recovery	Nominal	%	84.00	84.00	84.00	84.00	1	Owner	44	Range 80-88%	A
	Flotation Concentrate Leach Gold Recovery	Maximum	%	88.00	88.00	88.00	88.00	1	Owner	44	Range 80-88%	A
	Flotation Tails Leach Gold Recovery	Nominal	%	34.00	34.00	34.00	34.00	1	Owner	44	34% YTD	A
	Flotation Tails Leach Gold Recovery	Maximum	%	34.00	34.00	34.00	34.00	1	Owner	44	34% YTD	A
2.7	Gold Production											
	Nominal	oz Au/a		1,108,133	1,194,627	1,195,298	1,231,759	6	Calculated			
2.8	Flotation Concentrate Production	t/a		636,932	675,971	676,099	707,270	6	Calculated			
2.9	Process Water SG			1.070		1.070	1.070	1	Owner	1		A
2.10	Scheme Water SG			1.000		1.000	1.000	1	Owner	1		B
3.0	CRUSHING CIRCUIT											
3.1	Circuit Parameters											
	Throughput	Nominal	th	1,844	2,634	2,370	2,634	6	Calculated		New circuit	B
		Design	th		3,135	3,135	3,135	1	Owner	43		
		Maximum	th		3,436	3,436	3,436	1	Owner	56	Crusher fall through from FLS and not the circuit design basis	B
	Feed PSD											
	F ₁₀	mm		1,000	1,000	1,000	1,000	1	Owner	23		A
	F ₅₀	mm		450	450	450	450	1	Owner	23		A
	F ₈₀	mm		150	150	150	150	1	Owner	23		A
	F ₂₀₀	mm		25	25	25	25	1	Owner	23		A
	Product PSD											
	P ₈₀ - Maximum	mm	100	130	130	130	145	1, 5	Owner, Vendor	23, 25		A
3.2	Dump Pocket											
	Dump Pocket Type			Direct Tip, Two Sides	Direct Tip, Two Sides	Direct Tip, Two Sides	Direct Tip, Two Sides	1	Owner	16		A
	Truck Model			793F	793F	793F	793F	1	Owner	23		A
	Truck Payload	t		250	250	250	250	1	Owner	78	Owner's Manual CAT793F.	B
	SAE 2:1 Heaped Tray Capacity	m ³		179	179	179	179	1	Owner	76, 78	Owner's Manual CAT793F.	B
	Dump Pocket Capacity	m ³		179	179	179	179	1	Owner	76, 78	Live capacity from one side	B
3.3	Primary Crusher											
	Type		Gyratory	Gyratory	Gyratory	Gyratory	Gyratory	1	Owner	16	Owner Supplied	A
	Model		54-75Mk II	1300 x 1900 TSUV	1300 x 1900 TSUV	1300 x 1900 TSUV	1300 x 1900 TSUV	1	Owner	41	Owner Supplied	A
	Open Side Setting	mm	141	165	165	165	165	5	Vendor	51		B
	Throw	mm	38	38	38	38	38	1	Owner	42		B
	Motor Power	kW	450	600	600	600	600	5	Vendor	42		A
	Simulation Results											
	P ₅₀	Max	mm	122	122	122	122	5	Vendor	51	Simulated, see reference document for inputs	B
	P ₁₀₀	Max	mm	229	229	229	229	5	Vendor	51	Simulated, see reference document for inputs	B

AREA	PARAMETER	UNIT	DATA	REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4		
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um	
3.4	Crushed Ore Storage Type		Stockpile	Stockpile	Stockpile	Stockpile	Stockpile	1 Owner 16
	Live Capacity - Existing	t	21,001	-	-	-	-	6 Calculated
	Residence Time - Existing	h	16	-	-	-	-	6 Calculated
	Bulk Density	tm ³		1.60 - 1.80	1.60 - 1.80	1.60 - 1.80	1.60 - 1.80	1 Owner 43
	Live Capacity - New	m ³		12,035	12,035	12,035	12,035	2 EPC 37
	Residence Time - Minimum/Range	h		11.16	12.40	11.16	9.38	6 Calculated
	Live Capacity at 1.70 tm ³	t		20,460	20,460	20,460	20,460	6 Calculated
	Live Capacity at 1.90 tm ³	t		22,867	22,867	22,867	22,867	6 Calculated
	Live Capacity - New	m ³		11,293	11,293	11,293	11,293	2 EPC 37
	Residence Time - Minimum/Range	h		10.47	11.64	10.47	8.80	6 Calculated
	Live Capacity at 1.70 tm ³	t		19,198	19,198	19,198	19,198	6 Calculated
	Live Capacity at 1.90 tm ³	t		21,457	21,457	21,457	21,457	6 Calculated
	Primary Crusher Discharge Feeder Number			1	1	1	1	1 Owner 43
	Type			Apron	Apron	Apron	Apron	1 Owner 16
	Capacity	Nominal		2,634	2,634	2,634	3,135	6 Calculated
		Design		3,135	3,135	3,135	3,135	1 Owner 43
3.5	Reclaim Feeders Number		4	2	2	2	2	1,2 EPC Owner 16, 26
	Type		Apron	Apron	Apron	Apron	Apron	1 Owner 16
	Capacity per feeder	Nominal		917	917	917	1,091	6 Calculated
		Design		2,017	2,017	2,017	2,017	9 Agreed 53
3.6	Emergency Feeders Number		1	1	1	1	1	1 Owner 16
	Type		Apron	Apron	Apron	Apron	Apron	1 Owner 16
	Capacity per feeder	Nominal		1,833	1,833	1,833	2,182	6 Calculated
		Design		2,017	2,017	2,017	2,017	9 Agreed 53
	Loader Model			992K/994K	992K/994K	992K/994K	992K/994K	1 Owner 23
4.0	GRINDING AND CLASSIFICATION							
4.1	Circuit Parameters Type		SABC	SABC	SABC	SABC	SABC	1 Owner 16
	Nom Throughput	Nominal @ 180 µm	1,313	1,833	1,650	1,833		6 Calculated
	Nom Throughput	Nominal @ 220 µm					2,182	6 Calculated
	Bulk Density	Design Range		1.60 - 1.90	1.60 - 1.90	1.60 - 1.90	1.60 - 1.90	1 Owner 43, 62
	Milling Circuit PSD							
	P ₈₀		100	130	130	130	145	1,5 Owner, Vendor 23, 25
	P ₁₀₀	Target	180	180	180	180	220	1 Owner 16, 23, 24
	SAG Mill Specific Energy - 50th Percentile		8.3					1 Owner 16
	SAG Mill Specific Energy - 80th Percentile		8.9					1 Owner 16
	Ball Mill Specific Energy - 50th Percentile		5.4					1 Owner 16
	Ball Mill Specific Energy - 80th Percentile		6.1					1 Owner 16
	P ₅₀	Modelled		189	189	189	220	1 Owner 25
	SAG Mill Specific Energy Modelled			9.14	9.14	9.14	7.68	1 Owner 25
	Ball Mill Specific Energy Modelled	@ 180 µm		9.68	9.68	9.68		1 Owner 25
	Ball Mill Specific Energy Modelled	@ 220 µm					8.14	2 EPC 25
4.2	Primary Mill Type		SAG Mill	SAG Mill	SAG Mill	SAG Mill	SAG Mill	1 Owner 16
	Mill Diameter	m	10.97	10.97	10.97	10.97	10.97	1 Owner 16
	Effective Grinding Length - EGL	m	4.27	8.38	8.38	8.38	8.38	1 Owner 16, 23
	Duty Mill Speed Range	% Cs	60 - 80	60 - 83	60 - 83	60 - 83	60 - 83	1 Owner 24
	Duty Ball Charge	% v/v	13 - 17	10.5	10.5	10.5	10.9 - 11.3	1 Owner 16, 24
	Duty Total Load	% v/v	25 - 30	27	27	27	25	1 Owner 16, 24
	Total Ball Charge	Structural		18	18	18	18	1 Owner 23
	Total Load	Structural		35	35	35	35	1 Owner 23
	Mill Discharge Density	% w/w	75	75	75	75	75	1 Owner 9, 69
	Mill Pinion Power	Average	10,894	-	-	-	-	1 Owner 25
	Mill Pinion Power	Duty	11,682	16,078	15,078	16,757	16,755	6 Calculated
	Installed Motor Power		12,000	20,000	20,000	20,000	20,000	1 Owner 41, 23
	Liner Type			Steel	Steel	Steel	Steel	1 Owner 41
	Curved Pulp Lifter Capacity	Max		2,100	2,100	2,100	2,100	1 Owner 41
	Pebble Port Aperture		75	75	75	75	75	1 Owner 41
	Discharge Grate Open Area			11 - 12	11 - 12	11 - 12	11 - 12	1 Owner 41
	Trommel Dimensions	ID x L		3,829 m dia. X 3.80 m	3,829 m dia. X 3.80 m	3,829 m dia. X 3.80 m	3,829 m dia. X 3.80 m	5 Vendor 92
	Trommel Open Area	%	25	25	25	25	25	1 Owner 41
	Trommel Aperture	mm		12 x 40	12 x 40	12 x 40	12 x 40	1 Owner 41
	Spray Water	m ³ /h	150	150	150	150	150	2 EPC 40
	Trommel Undersize	t/h		1,050	1,050	1,050	1,050	2 EPC 32
	Existing Fim SAG Mill Transfer Density	% w/w	60					1 Owner 57, 69
4.3	Primary Mill Discharge Screen Discharge Screen Type			Horizontal Vibrating Screen	Horizontal Vibrating Screen	Horizontal Vibrating Screen	Horizontal Vibrating Screen	1 Owner 16
	Screen Size	W x L		4.27 x 8.54	4.27 x 8.54	4.27 x 8.54	4.27 x 8.54	5 Vendor 59
	Effective Screen Area	m ²		36.47	36.47	36.47	36.47	5 Vendor 59
	Screen Open Area	%		34.0	34.0	34.0	34.0	5 Vendor 59
	Screen Aperture	mm		12 x 55	12 x 55	12 x 55	12 x 55	2 EPC 40
	Specific Screen Flux Rate	m ³ /m ² /h OA		18.68	18.68	18.68	29.90	5 Vendor 59
	Oversize Moisture	%w/w	5	13	13	13	13	5 Vendor 59
	Spray/Wash Water	m ³ /h		197.2	197.2	197.2	197.2	5 Vendor 59
	Discharge Screen Feed % Solids	%w/w		85	85	85	85	2 EPC 32
	Discharge Screen Oversize	t/h		375	375	375	375	1 Owner 41
4.4	Secondary Mill Type			Overflow Ball Mill	Overflow Ball Mill	Overflow Ball Mill	Overflow Ball Mill	1 Owner 41
	Mill Diameter	m		8.53	8.53	8.53	8.53	1 Owner 23
	Effective Grinding Length - EGL	m		13.26	13.26	13.26	13.26	1 Owner 23
	Duty Mill Speed	% Cs		60 - 78	60 - 78	60 - 78	60 - 78	1 Owner 41
	Duty Ball Charge	Nominal		28.5	28.5	28.5	28.5	1 Owner 41
	Duty Ball Charge	Maximum		31.0	31.0	31.0	31.0	1 Owner 24
	Total Ball Charge	Structural		38	38	38	38	1 Owner 23
	Mill Discharge Density	% w/w		75	75	75	75	1 Owner 4
	Mill Pinion Power	Duty		17,747	15,969	17,747	17,759	6 Calculated
	Installed Motor Power			20,000	20,000	20,000	20,000	1 Owner 23
	Liner Type			Poly-Met	Poly-Met	Poly-Met	Poly-Met	1 Owner 23
	Trommel Flux Rate	Nominal		223	223	223	226	6 Calculated
	Trommel Dimensions	ID x L		4,992 x 5.80	4,992 x 5.80	4,992 x 5.80	4,992 x 5.80	5 Vendor 92
	Trommel Capacity	Maximum		6,681	6,681	6,681	6,681	1 Owner 23
	Mill Discharge Flowrate	Nominal		6,373	6,197	6,383	6,277	6 Calculated
	Trommel Open Area	Minimum		30	30	30	30	1 Owner 23
	Trommel Aperture	mm		19 x 40 - slotted	19 x 40 - slotted	19 x 40 - slotted	19 x 40 - slotted	1 Owner 23
	Trommel Screen Spray Water	m ³ /h		150	150	150	150	1 Owner 4
4.5	Pebble Crusher Type		Cone	Cone	Cone	Cone	Cone	1 Owner 16
	Number		2 (duty / standby)	2 (duty / duty)	2 (duty / duty)	2 (duty / duty)	2 (duty / duty)	1 Owner 16, 23
	Model		Symons 78	XL7 SH-HD	XL7 SH-HD	XL7 SH-HD	XL7 SH-HD	1 Owner 41
	Installed Motor Power per Crusher	kW	375 / 260	375	375	375	375	1 Owner 16, 41, 44
	Closed Side Setting	mm	7 - 9	7	7	7	7	1 Owner 16, 41
	Pebble Crushing Work Index	Maximum		23	23	23	23	1 Owner 56
	Pebble Bulk Density	Range		1.6 - 1.9	1.6 - 1.9	1.6 - 1.9	1.6 - 1.9	1 Owner 56
	Throughput	Nominal		302				6 Calculated
	Throughput	Nominal		302				1 Owner 56
	Range	t/h		305	305	305	610	1 Owner 41
	% of new feed	% of new feed		23 - 27				1 Owner 16
	% of new feed	% of new feed		23.0				1 Owner 16
	Crusher Capacity at CSS	Maximum		16.6	18.5	16.6	28.0	6 Calculated
	Feed Bin Capacity	per bin		610	610	610	610	1 Owner 41
	Feed Size F ₅₀	Typical						
	Feed Size F ₁₀₀	Modelled		276	276	276	276	1 Owner 16
	Feed Size F ₅₀	Modelled		350	350	350	350	1 Owner 9
	Feed Size F ₅₀	Modelled		0.757	0.757	0.757	0.757	1 Owner 9
	Feed Size F ₅₀	Modelled		0.306	0.306	0.306	0.306	1 Owner 9
	Product Size P ₁₀	Target		180	180	180	220	1 Owner 26
	Product Size P ₅₀	Maximum		250	250	250	250	1 Owner 26
	Type			Hydrocyclone	Hydrocyclone	Hydrocyclone	Hydrocyclone	1 Owner 16
	Cyclone Model			800CVX13	800CVX13	800CVX13	800CVX13	5 Vendor 28
	Number of Cyclone Clusters			1	1	1	1	5 Vendor 28
	Distributor Outlets per Cluster	No.		18	18	18	18	5 Vendor 28
	Number Installed Cyclones	No.		17	17	17	17	5 Vendor 28
	Number Spare Outlets	No.		1	1	1	1	6 Calculated
	Number Operating Cyclones - Primary Cluster	Nominal		12	11	12	12	5 Vendor 28
		Maximum		14	14	14	14	5 Vendor 28
		Standby		1	1	1	1	5 Vendor 28
	Total Installed	No.		15	15	15	15	6 Calculated

AREA	PARAMETER		UNIT		DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV	
					1	2	3	4						
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um						
4.7	Number Operating Cyclones - Gravity	Nominal	No.		2	2	2	2	5	Vendor	28	Gravity cyclones require larger spigot and VF to achieve mass flow of 1000tph to underflow. There is also a higher recirc. load.	A	
		Maximum	No.		2	2	2	2	5	Vendor	28		A	
	Cyclone Feed	Nominal	th		14,421	13,986	14,421	14,421	6	Calculated				
	Cyclone Feed Flow	Nominal	m³/h		13,233	12,816	13,217	13,217	6	Calculated				
	Cyclone Feed Flow	Maximum @ 61.6%	m³/h		13,862	13,862	13,862	13,862	1	Owner	26		Flow rate of 13,862m³/hr requires 14 or 15 online. 2 gravity as advised by Weir	A
	Cyclone Feed Density		% w/w		62.0	62.0	62.0	65.0	5	Vendor	28			A
	Underflow Solids Concentration	Minimum	% w/w		75.0	75.0	75.0	77.2	1	Owner	16			B
	Underflow Solids Concentration		% w/w		77.2	77.0	77.2	77.2	6	Calculated				
	Underflow Solids		th		11,275	11,024	11,275	11,129	6	Calculated				
	Overflow Solids Concentration		% w/w		36.4	35.9	36.4	43.3	5	Vendor	28			A
	Overflow Solids		th		3,146	2,962	3,146	3,494	6	Calculated				
	Circulating Load - Cyclone UF / Cyclone OF		%		358	372	358	318	5	Vendor	28		A	
	Circulating Load - Ball Mill Discharge / Cyclone OF		%		301	311	301	267	6	Calculated				
	Operating Pressure - Cyclones in Cluster		kPa		82	91	82	73	5	Vendor	28		B	
	Operating Pressure - Gravity Cyclones		kPa		83	86	83	70	5	Vendor	28		B	
	Trash Screen													
	Type				Linear Screen	Linear Screen	Linear Screen	Linear Screen	1	Owner	16		A	
	Total Number Screens				4	4	4	4	6	Calculated				
	Number Operating				3	3	3	3	1	Owner	14		A	
	Number Standby				1	1	1	1	1	Owner	14		A	
	Screen Dimensions	W x L	m		9.53 x 4.2	9.53 x 4.2	9.53 x 4.2	9.53 x 4.2	5	Vendor	55		B	
	Screen Area per Screen	Effective	m²		40.03	40.03	40.03	40.03	5	Vendor	55		A	
	Total Screen Area		m²		120.09	120.09	120.09	120.09	6	Calculated				
	Effective Screen Area		%		34	34	34	34	5	Vendor	55		B	
	Screen Aperture		mm		1.0 x 1.0	1.0 x 1.0	1.0 x 1.0	1.0 x 1.0	5	Vendor	55		B	
	Screen Feed Flow (incl. spray water)		m³/h		2,183	2,093	2,182	1,936	6	Calculated				
	Maximum Screen Flux Rate		m³/m²/h		75	75	75	75	1	Owner	26		A	
	Specific Screen Flux Rate		m³/m²/h		54.5	52.3	54.5	48.4	6	Calculated				
	Spray/Wash Water	per screen	m³/h		60	60	60	60	5	Vendor	31, 55		B	
	Total Spray/Wash Water		m³/h		180	180	180	180	6	Calculated				
	Spray Water reporting to Screen Oversize	per screen	m³/h		45	45	45	45	5	Vendor	31		B	
	Total Spray Water reporting to Screen Oversize	Total	m³/h		135	135	135	135	6	Calculated				
	Spray Water reporting to Screen Undersize	per screen	m³/h		15	15	15	15	5	Vendor	31		B	
	Spray Water reporting to Screen Undersize	Total	m³/h		45	45	45	45	6	Calculated				
	Water to Trash Screens Final Collection Box		m³/h		258.0	134.9	258.0	306.8	6	Calculated		HP Process Water, back calculated in SysCAD	B	
4.8	Trash Dewatering													
	Dewatering Screen Type			Vibrating Screen	Vibrating Screen	Vibrating Screen	Vibrating Screen	1	Owner	16		A		
	Number			1	1	1	1	1	Owner	16		A		
	Screen Size	W x L	m	1.83 x 3.66	1.83 x 3.66	1.83 x 3.66	1.83 x 3.66	5	Vendor	59		B		
	Effective Screening Area per Screen		m²	6.14	6.14	6.14	6.14	5	Vendor	59		B		
	Specific Screen Flux Rate	Design	m³/m²/h	50	50	50	50	1	Owner	16		A		
		Nominal	m³/m²/h	22.36	22.36	22.36	22.36	5	Vendor	59				
	Screen Aperture		mm	1 x 15	1 x 15	1 x 15	1 x 15	1	Owner	16		A		
	Deck Type			Polyurethane	Polyurethane	Polyurethane	Polyurethane	5	Vendor	59		A		
	Quantity of Wood/Grit/Trash removed per day	td/day		27	27	27	27	1	Owner	16				
	Total Spray/Wash Water		m³/h	11	11	11	11	5	Vendor	59		B		
5.0 GRAVITY CONCENTRATION														
5.1 Gravity Scalping Screen														
5.1	Type			Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	5	Vendor	59		A		
	Number of Screens			2	2	2	2	1	Owner	16				
	Screen Solids Feed Rate per Screen	Nominal	th	1,000	1,000	1,000	1,000	6	Calculated			Only 2x gravity concentrator circuits, no future allowance for additional circuits	A	
		Design	th	1,111	1,111	1,111	1,111	6	Calculated			To achieve 1,000 t/h to gravity concentrators at design O/S split	A	
	Screen Feed Density	Target	% w/w	60.0	60.0	60.0	60.0	1	Owner	70		Excludes Spray Water	B	
	Screen Feed Density	Actual	% w/w	60.0	60.0	60.0	60.0	6	Calculated					
	Screen Feed Flowrate per Screen	Nominal	m³/h	968	966	967	967	6	Calculated			Excludes Spray Water		
		Design	m³/h	1,075	1,073	1,074	1,075	6	Calculated			Excludes Spray Water		
	Screen Size	W x L	m	3.66 x 6.10	3.66 x 6.10	3.66 x 6.10	3.66 x 6.10	1	Owner	59, 70, 72			B	
	Effective Screen Area per Screen	Overall	m²	22.31	22.31	22.31	22.31	5	Vendor	59				
	Specific Screen Flux Rate	Nominal	m³/m²/h	48.0	47.9	47.9	47.9	6	Calculated			Includes Spray Water		
		Design	m³/m²/h	52.8	52.7	52.7	52.7	6	Calculated			Includes Spray Water		
		Target	m³/m²/h	50	50	50	50	1	Owner	16, 70			A	
	Screen Split to Oversize - Range		%	8 - 15	8 - 15	8 - 15	8 - 15	1	Owner	16			A	
		Design	%	10	10	10	10	1	Owner	16, 72			A	
	Screen Aperture		mm	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	5	Vendor	59		3.2mm Slotted first half, 2mm non-pegging second half (VR) requested by owner in source 16	B	
	Screen Spray Water	per screen	m³/h	102	102	102	102	5	Vendor	59			B	
	Oversize Solids Concentration		% w/w	87	87	87	87	5	Vendor	59			B	
	Oversize Slurry Destination			Ball mill feed	Ball mill feed	Ball mill feed	Ball mill feed	1	Owner	43			A	
	Undersize Slurry Destination			Gravity Concentrators	Gravity Concentrators	Gravity Concentrators	Gravity Concentrators	1	Owner	43			A	
5.2	Gravity Concentrator													
	Type			Knelson	Knelson	Knelson	Knelson	1	Owner	16			A	
	Number of Concentrators			2	2	2	2	1	Owner	16		Only 2x gravity concentrator circuits	A	
	Model			KC-QS70	KC-QS70	KC-QS70	KC-QS70	1	Owner	16			A	
	Feed Capacity per Unit	Max	th	1,000	1,000	1,000	1,000	5	Vendor	39, 71			A	
	Volumetric Capacity per Unit	Max	m³/h	910	910	910	910	5	Vendor	7			B	
	Feed Solids Rate to Circuit		th	1,800	1,800	1,800	1,800	2, 5	EPC Vendor	7, 71		300-1000t/hr/unit operating range	A	
	Feed Solids Rate per Unit	Max	th	2,000	2,000	2,000	2,000	6	Calculated					
	Feed Solids Rate per Unit	Nominal	th	900	900	900	900	6	Calculated			Vendor recommends 90% of the max (Source 71).		
	Feed Density		% w/w	54.2	54.2	54.2	54.2	6	Calculated					
	Volumetric Flowrate per Unit		m³/h	1,021	1,020	1,020	1,021	6	Calculated					
	Fluidising Water per Unit		m³/h	120	120	120	120	5	Vendor	7			B	
	Fluidising Water	Nominal	m³/h	240	240	240	240	6	Calculated			120m³ per concentrator		
	Cycle Time (Inclusive of Flush Cycle)		min	30	30	30	30	5	Vendor	71			B	
	Cycles Per Day		No.	48	48	48	48	6	Calculated					
	Concentrate Production Per Cycle		kg	150	150	150	150	5	Vendor	7, 71			A	
	Concentrate Bulk SG		t/m³	2.20	2.20	2.20	2.20	1	Owner	44			B	
Daily Concentrate Production		kg/d	14,400	14,400	14,400	14,400	6	Calculated						
		m³/d	6.55	6.55	6.55	6.55	6	Calculated						
Fresh Feed to Concentrators		%	57.2	60.8	57.2	51.5	6	Calculated						
Tailings Destination			Cyclone Feed Hopper	Cyclone Feed Hopper	Cyclone Feed Hopper	Cyclone Feed Hopper	1	Owner	43			A		
6.0 INTENSIVE LEACH														
6.1 Intensive Leach Reactor														
6.1	Type			Acacia	Acacia	Acacia	Acacia	1	Owner	16			A	
	Model			CS10000	CS10000	CS10000	CS10000	1	Owner	70			B	
	Number			1	1	1	1	1	Owner	16			A	
	Batches per Day	Nominal		1	1	1	1	5	Vendor	39			A	
		Max		2	2	2	2	5	Vendor	39		Required to complete 2 batches per day as per Owner Source 72	A	
	Solids Feed Rate per Batch		kg/batch	14,400	14,400	14,400	14,400	6	Calculated					
	Design Solids Feed Rate per Batch		kg/batch	15,000	</									

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
7.0	FLOTATION										
7.1	General										
	Throughput	t/h		3,146	2,962	3,146	3,494	6	Calculated		
	Feed Particle Size - p80	µm		180	180	180	220	1	Owner	26	A
		Design		220	220	220	220	1	Owner	26	B
	Froth Factors										
	Rougher Flotation Concentrate			2.00	2.00	2.00	2.00	1	Owner	16	A
	Scavenger Flotation Concentrate			1.60	1.60	1.60	1.60	1	Owner	16	A
	Cleaner Flotation Concentrate			2.50	2.50	2.50	2.50	1	Owner	16	A
	Cleaner Scavenger Flotation Concentrate			2.00	2.00	2.00	2.00	1	Owner	16	A
	Flotation Cell Transfer Lines			1.20	1.20	1.20	1.20	1	Owner	16	A
	Launders			-	-	-	-	1	Owner	16	A
	Spray Water Line Pressure (at vendor battery limit)	kPa		62.74	62.74	62.74	62.74	5	Vendor	60	B
7.2	Rougher Flotation										
	Feed Streams			Trash Screen Underflow	Trash Screen Underflow	Trash Screen Underflow	Trash Screen Underflow	1	Owner	43	Owner Supplied
	Solids Feed Rate	New Feed	t/h	3,146	2,962	3,146	3,494	6	Calculated		A
		Total	t/h	3,161	2,977	3,160	3,512	6	Calculated		
	Volumetric Feed Rate		m³/h	6,719	6,326	6,716	6,027	6	Calculated		Includes sample return
	Feed Solids Concentration		% w/w	34.4	34.4	34.4	40.5	6	Calculated		
	Particle Size p80	Design	µm	220	220	220	220	2	EPC	63	Feed, concentrate and tailings p80 assumed equal (conservative)
	Mass Pull		%	4.4	4.4	4.4	4.4	6	Calculated		
	Concentrate 1 Upgrade Ratio			15	15	15	15	1	Owner	16	A
	Concentrate 1 Grade		% S	15.3	22.4	21.0	15.3	6	Calculated		
	Concentrate 1 Stage Recovery		% Au in fl. feed	65	65	65	65	1	Owner	23	A
	Concentrate 1 Stage Recovery		% of S in fl. feed	65	65	65	65	1	Owner	23	A
	Concentrate 1 Production		t/h	138	130	138	153	6	Calculated		
	Concentrate 1 Solids Concentration		% w/w	25	25	25	25	1	Owner	16	A
	Concentrate 1 Slurry Volume - Excl Spray Water		m³/h	427	400	425	474	6	Calculated		
	Concentrate 1 SG			3.32	3.54	3.50	3.32	2	EPC	32	A
	Concentrate 1 Launder Spray Water	Operating Range	m³/h	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60	B
		Design	m³/h	50	50	50	50	5	Vendor	60	A
	Mass Pull		%	2.4	2.4	2.4	2.4	6	Calculated		
	Concentrate 2 & 3 Upgrade Ratio			7	7	7	7	1	Owner	16	A
	Concentrate 2 & 3 Grade		% S	7.1	10.4	9.8	7.1	6	Calculated		
	Concentrate 2 & 3 Stage Recovery		% Au in fl. feed	17	17	17	17	1	Owner	23	A
	Concentrate 2 & 3 Stage Recovery		% of S in fl. feed	17	17	17	17	1	Owner	23	A
	Concentrate 2 & 3 Production		t/h	77	72	77	86	6	Calculated		
	Concentrate 2 & 3 Solids Concentration		% w/w	25	25	25	25	1	Owner	16	A
	Concentrate 2 & 3 Slurry Volume - Excl Spray Water		m³/h	241	225	239	268	6	Calculated		
	Concentrate 2 & 3 SG			3.06	3.17	3.15	3.06	2	EPC	32	A
	Concentrate 2 Launder Spray Water	Operating Range	m³/h	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60	
	Concentrate 3 Launder Spray Water	Operating Range	m³/h	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60	
	Concentrate 2 & 3 Spray Water	Design (Combined)	m³/h	50	50	50	50	5	Vendor	50	A
	Rougher Tail Solids Rate		t/h	2,946	2,775	2,946	3,273	6	Calculated		
	Rougher Tail Slurry Volumetric Flow		m³/h	6,051	5,701	6,052	5,285	6	Calculated		
	Cell Type			Tank Cells	Tank Cells	Tank Cells	Tank Cells	1	Owner	16	A
	Aspiration			Forced	Forced	Forced	Forced	1	Owner	16	A
	Number Cells Installed			3	3	3	3	1	Owner	41	A
	Nominal Cell Volume		m³	630	630	630	630	1	Owner	16	A
	Total Cell Volume		m³	1,890	1,890	1,890	1,890	6	Calculated		
	Cell Air Hold-Up		%	15	15	15	15	1	Owner	41	A
	Plant Residence Time		min	15.1	16.0	15.1	17.0	6	Calculated		Rougher residence time existing plant = 10-12 mins (source 23). Residence time has been calculated using the average of the feed and tailings flowrates.
	Lip Loading - Design		t/m/h	1.5	1.5	1.5	1.5	1	Owner	16	A
	Froth Carry Rate - Design		t/m²/h	1.5	1.5	1.5	1.5	1	Owner	16	A
7.3	Scavenger Flotation										
	Feed Streams			Rougher Tailings	Rougher Tailings	Rougher Tailings	Rougher Tailings	1	Owner	43	Owner Supplied
	Solids Feed Rate		t/h	2,946	2,775	2,946	3,273	6	Calculated		A
	Volumetric Feed Rate		m³/h	6,051	5,701	6,052	5,286	6	Calculated		
	Feed Solids Concentration		% w/w	35.3	35.3	35.3	42.4	6	Calculated		
	Particle Size p80	Design	µm	220	220	220	220	2	EPC	63	Feed, concentrate and tailings p80 assumed equal (conservative)
	Mass Pull (Scavenger Concentrate)		%	1.1	1.1	1.1	1.1	6	Calculated		
	Scavenger Concentrate Upgrade Ratio			5	5	5	5	1	Owner	16	A
	Scavenger Concentrate Grade		% S	5.1	7.5	7.0	5.1	6	Calculated		
	Scavenger Concentrate Stage Recovery		% Au in fl. feed	5	5	5	5	1	Owner	23	A
	Scavenger Concentrate Stage Recovery		% of S in fl. feed	5	5	5	5	1	Owner	23	A
	Scavenger Concentrate Production		t/h	32	30	32	35	6	Calculated		
	Scavenger Concentrate Solids Concentration		% w/w	25	25	25	25	2	EPC	32	A
	Scavenger Concentrate Slurry Volume - Excl Spray Water		m³/h	99	93	99	110	6	Calculated		
	Scavenger Concentrate SG			3.00	3.07	3.06	3.00	2	EPC	32	A
	Scavenger Concentrate Solids Concentration - Diluted		% w/w	20	20	20	20	2	EPC	32	A
	Scavenger Concentrate Launder Spray Water	Actual (Combined)	m³/h	29	28	29	33	6	Calculated		
		Operating Range	m³/h/cell	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60	B
		Design	m³/h/cell	25.00	25.00	25.00	25.00	5	Vendor	60	B
	Scavenger Tail Solids Rate		t/h	2,914	2,745	2,914	3,238	6	Calculated		
	Scavenger Tail Slurry Volumetric Flow		m³/h	5,952	5,608	5,953	5,176	6	Calculated		
	Cell Type			Tank Cells	Tank Cells	Tank Cells	Tank Cells	1	Owner	16	A
	Aspiration			Forced	Forced	Forced	Forced	1	Owner	16	A
	Number Cells Installed			4	4	4	4	1	Owner	41	A
	Nominal Cell Volume		m³	630	630	630	630	1	Owner	16	A
	Total Cell Volume		m³	2,520	2,520	2,520	2,520	6	Calculated		
	Cell Air Hold-Up		%	15	15	15	15	1	Owner	41	A
	Plant Residence Time		min	21.4	22.7	21.4	24.6	6	Calculated		Scavenger residence time existing plant = 29 - 32 mins (source 23). Residence time has been calculated using the average of the feed and tailings flowrates.
	Lip Loading - Design		t/m/h	1.5	1.5	1.5	1.5	1	Owner	16	A
	Froth Carry Rate - Design		t/m²/h	1.5	1.5	1.5	1.5	1	Owner	16	A
	Combined Rougher and Scavenger Residence Time			36.5	38.8	36.5	41.6	6	Calculated		
7.4	Cleaner Flotation										
	Feed Streams			Rougher and Scavenger Concentrate	Rougher and Scavenger Concentrate	Rougher and Scavenger Concentrate	Rougher and Scavenger Concentrate	1	Owner	43	Owner Supplied
	Solids Feed Rate	New Feed Only	t/h	253	244	258	282	6	Calculated		New feed only
		Total	t/h	465	424	439	478	6	Calculated		Total feed to Jameson including tails recycle and sample return minus cleaner feed sample to OSA
	Volumetric Feed Rate to Cell - Target	Total	m³/h	1,998	1,998	1,998	1,998	5	Vendor	27	Average of four modelled cases
	Volumetric Feed Rate to Cell - Actual	Total	m³/h	1,998	1,998	1,998	1,998	6	Calculated		Total feed to Jameson including tails recycle and sample return minus cleaner feed sample to OSA
	Feed Solids Concentration - Actual	Total	% w/w	19.0	17.5	18.0	19.5	6	Calculated		Total feed to Jameson including tails recycle and sample return minus cleaner feed sample to OSA
	Particle Size p80	Design	µm	220	220	220	220	6	Calculated		Feed, concentrate and tailings p80 assumed equal (conservative).
		Calculated	µm	180	180	180	220	6	Calculated		Max of calculated p80
	Mass Pull		%	30.8	44.0	41.5	30.7	6	Calculated		
	Concentrate Grade		% S	35	35	35	35	1	Owner	26	A
	Concentrate Stage Recovery		% S	93.3	93.6	93.4	93.1	6	Calculated		
	Concentrate Recovery	Overall	% of S in fl. feed	85.1	85.1	85.1	85.1	1	Owner	23	A
	Concentrate Recovery	Overall	% Au in fl. feed	84	83	83	84	6	Calculated		
	Concentrate Production		t/h	78	107	107	86	6	Calculated		
	Concentrate Solids Concentration		% w/w	25	25	25	25	5	Vendor	27	Undiluted cleaner flotation concentrate
	Concentrate Slurry Volume - Excl Spray Water		m³/h	237	326	326	263	6	Calculated		
	Concentrate SG			3.93	3.93	3.93	3.93	2	EPC	32	A
	Concentrate Wash Water Ratio (of Con m3/hr)			1.0	1.0	1.0	1.0	5	Vendor	27	A
	Cleaner Tail Solids Rate		t/h	183	145	159	203	6	Calculated		To Cleaner-Scavenger Flotation Cell Feed Hopper
	Cleaner Tail Slurry Volumetric Flow		m³/h	945	910	957	1,038	6	Calculated		To Cleaner-Scavenger Flotation Cell Feed Hopper
	Cleaner Tails Recycle		%	52.7	54.4	52.1	48.1	6	Calculated		The % of total tailings recycled back to feed hopper.
	Cleaner Tail Solids Rate		t/h	204	173	173	188	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper
	Cleaner Tail Slurry Volumetric Flow		m³/h	1,053	1,068	1,041	960	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper
	Cell Type			Jameson	Jameson	Jameson	Jameson	5	Vendor	18	A
	Model			B6500/24	B6500/24	B6500/24	B6500/24	5	Vendor	27	A
	Aspiration Type			Natural	Natural	Natural	Natural	5	Vendor	16	A
	Number Cells Installed			1	1	1	1	5	Vendor	18	A
	Number of Downcomers			24	24	24	24	5	Vendor	27	B
	Froth Area		m²	22.2	22.2	22.2	22.2	5	Vendor	27	A
	Froth Carry Rate	Nom	t/m²/h	3.50	4.81	3.50	3.51	5	Vendor	27	B
		Max	t/m²/h	6.5	6.5	6.5	6.5	5	Vendor	27	A
		Calculated	t/m²/h	3.51	4.82	4.82	3.89	6	Calculated		

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV		
				1	2	3	4						
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um						
7.5	Cleaner Scavenger Flotation			Cleaner Flotation Tail	Cleaner Flotation Tail	Cleaner Flotation Tail	Cleaner Flotation Tail	1	Owner	66	Owner Supplied	B	
	Feed Streams			183	145	159	203	6	Calculated		New feed only		
	Solids Feed Rate	New Feed Only	t/h	380	300	317	385				Total feed to Jameson cell including tails recycle		
		Total	t/h	2,048	2,048	2,048	2,048	5	Vendor	27	Average of four modelled cases	C	
	Volumetric Feed Rate to Cell - Target	Total	m³/h	2,048	2,048	2,048	2,048	6	Calculated		Total feed to Jameson cell including tails recycle		
	Feed Solids Concentration	Total	% w/w	15.6	12.6	13.2	15.6	6	Calculated		Total feed to Jameson cell including tails recycle		
	Particle Size p80	Design	µm	220	220	220	220	2	EPC	63	Feed, concentrate and tailings p80 assumed equal (conservative). Max of calculated p80		
	Mass Pull		%	7.6	13.3	12.1	7.6	6	Calculated				
	Concentrate Grade		% S	13.10	13.10	13.10	13.10	5	Vendor	19	Design Grade sulphur grade based on EPC circuit modelling	A	
	Concentrate Stage Recovery		% S	49.6	54.1	54.2	50.9	6	Calculated			A	
	Concentrate Recovery		% of Au in fl. feed	5.7	5.7	5.7	5.7	3	Assumed		Assumed to match the S recovery as seen in other areas of the flotation circuit	A	
	Concentrate Recovery		% of S in fl. feed	5.7	5.7	5.7	5.7	2	EPC	32		A	
	Concentrate Production		t/h	14	19	19	15	6	Calculated				
	Concentrate Solids Concentration		% w/w	25	25	25	25	2	EPC	32		B	
	Concentrate Slurry Volume - Excl Spray Water		m³/h	43	60	60	48	6	Calculated				
	Concentrate SG			3.25	3.25	3.25	3.25	2	EPC	32		A	
	Concentrate Wash Water Ratio (of Con m3/hr)			1	1	1	1	5	Vendor	27		A	
	Cleaner Scavenger Tail Solids Rate		t/h	169	125	140	188	6	Calculated		To Cleaner-Scavenger Flotation Cell Tails Hopper		
	Cleaner Scavenger Tail Slurry Volumetric Flow		m³/h	948	914	960	1,041	6	Calculated		To Cleaner-Scavenger Flotation Cell Tails Hopper		
	Cleaner Scavenger Tails Recycle		%	53.7	55.4	53.1	49.2	6	Calculated		The % of total tailings recycled back to feed hopper.		
	Cleaner Scavenger Tail Solids Rate		t/h	196	159	182	192	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper		
	Cleaner Scavenger Tail Slurry Volumetric Flow		m³/h	1,100	1,134	1,088	1,007	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper		
	Cell Type			Jameson	Jameson	Jameson	Jameson	5	Vendor	19		A	
	Model			B6500/24	B6500/24	B6500/24	B6500/24	1	Owner	27		A	
	Aspiration Type			Natural	Natural	Natural	Natural	1	Owner	16		A	
	Number Cells Installed			1	1	1	1	5	Vendor	19		A	
	Number of Downcomers			24	24	24	24	5	Vendor	27		B	
	Froth Area		m²	22.2	22.2	22.2	22.2	5	Vendor	27		A	
	Froth Carry Rate	Nom	t/m²/h	0.63	0.85	0.63	0.63	5	Vendor	27		B	
		Max	t/m²/h	6.5	6.5	6.5	6.5	5	Vendor	27		A	
		Calculated	t/m²/h	0.63	0.86	0.86	0.70	6	Calculated				
	8.0	CONCENTRATE HANDLING											
	8.1	Concentrate Thickener (Brownfields)			Existing Flotation Tail Pre-Leach Thickener (CIL3)	Existing Flotation Tail Pre-Leach Thickener (CIL3)	Existing Flotation Tail Pre-Leach Thickener (CIL3)	Existing Flotation Tail Pre-Leach Thickener (CIL3)				Existing 30-TH-61	
		Feed Stream			Final Concentrate Hopper (Cleaner Flotation Concentrate)	Final Concentrate Hopper (Cleaner Flotation Concentrate)	Final Concentrate Hopper (Cleaner Flotation Concentrate)	Final Concentrate Hopper (Cleaner Flotation Concentrate)	1	Owner	16		A
		Solids Feed Rate		t/h	75	104	104	84	6	Calculated			
		Feed Flowrate - Excl. Diluted Flocculant		m³/h	230	319	319	256	6	Calculated			
Feed Solids Concentration			% w/w	24.9	24.9	24.9	24.9	6	Calculated				
Underflow Solids Concentration			% w/w	60	60	60	60	1	Owner	16		A	
Testwork Flux Rate			t/m²/h	0.25	0.25	0.25	0.25	1	Owner	23		A	
Calculated Flux Rate			t/m²/h	0.13	0.18	0.18	0.15	6	Calculated				
Thickener Area Calculated			m²	573	573	573	573	6	Calculated				
Diameter		Existing	m	27	27	27	27	1	Owner	16		A	
Structural Design SG				3.8	3.8	3.8	3.8	1	Owner	16		A	
Concentrate Thickener Sprays - Mill Circuit Water			m³/h	68.40	68.40	68.40	68.40	5	Vendor	113		1	
8.2	Filtered Feed Tanks A & B (Brownfields)			Existing	Existing	Existing	Existing						
	Feed Streams			Concentrate Thickener Underflow	Concentrate Thickener Underflow	Concentrate Thickener Underflow	Concentrate Thickener Underflow	1	Owner	43	Shared volume between Filtered Feed Tank A & B via balance line.	A	
	Solids Feed Rate		t/h	75	104	104	84	6	Calculated				
	Volumetric Feed Rate		m³/h	66	92	92	74	6	Calculated				
	Feed Solids Concentration		% w/w	59.6	59.7	59.7	59.6	6	Calculated				
	Number of Tanks			2	2	2	2	1	Owner	16		A	
	Residence Time		h	21.2	15.3	15.3	19.0	6	Calculated				
	Tank Capacity	per tank	m³	700	700	700	700	1	Owner	16		A	
	Concentrate Filters (Brownfields)			Existing	Existing	Existing	Existing				Remains in operation to treat additional concentrate in excess of the UFG capacity.		
	Feed Stream			Concentrate Filtered Feed Tank B	Concentrate Filtered Feed Tank B	Concentrate Filtered Feed Tank B	Concentrate Filtered Feed Tank B	1	Owner	43		A	
	Filter Type			Vacuum Belt Filter	Vacuum Belt Filter	Vacuum Belt Filter	Vacuum Belt Filter	1	Owner	43		A	
	Number of Filters			2	2	2	2	1	Owner	16		A	
Solids Feed Rate		t/h	0	0	0	0	6	Calculated		Solids have been prioritised through UFG circuits first.			
Volumetric Feed Rate		m³/h	0	0	0	0	6	Calculated					
Feed Solids Concentration	Target	% w/w	60.0	60.0	60.0	60.0	1	Owner	16		A		
	Actual	% w/w	0.0	0.0	0.0	0.0	6	Calculated					
Target Moisture Content		<% w/w	10.0	10.0	10.0	10.0	1	Owner	16		A		
Filter Area	per filter	m²	55.0	55.0	55.0	55.0	1	Owner	16		A		
Specific Filtration Rate	Nominal	m³/m²/h	0.14	0.14	0.14	0.14	1	Owner	16		A		
	Design	m³/m²/h	0.31	0.31	0.31	0.31	1	Owner	16		A		
8.4	Concentrate Stockpile			Concentrate Filter Cake	Concentrate Filter Cake	Concentrate Filter Cake	Concentrate Filter Cake				Remains in operation to treat additional concentrate in excess of the UFG capacity.		
	Feed Stream			1	1	1	1	1	Owner	43		A	
	No. of Stockpiles			1	1	1	1	1	Owner	16		A	
	Concentrate Bulk Density		t/m³	2.2	2.2	2.2	2.2	1	Owner	16		A	
	Angle of Repose	Nominal	Degrees	40.0	40.0	40.0	40.0	1	Owner	16		A	
	Storage		days	2	2	2	2	1	Owner	16		A	
	Capacity		m³	1,139	1,139	1,139	1,139	1	Owner	16		A	
	TML for Bulk Shipment		% w/w	6.10 - 8.26	6.10 - 8.26	6.10 - 8.26	6.10 - 8.26	1	Owner	23		A	
	9.0	CONCENTRATE ULTRAFINE GRINDING											
	9.1	Concentrate UFG Mill 1 (Brownfields)			Existing	Existing	Existing	Existing				M3000	
		Circuit Feed Streams			Concentrate Filter Feed Tank A	Concentrate Filter Feed Tank A	Concentrate Filter Feed Tank A	Concentrate Filter Feed Tank A	1	Owner	16, 80		1
		Feed Rate - Fresh Feed	P80 9 µm	t/h	0.0	10.0	10.0	0.0	1	Owner	69		B
		Design	t/h	0.0	13.0	13.0	0.0	1	Owner	16		A	
Feed Solids Concentration			% w/w	45	45	45	45	1	Owner	4		A	
Feed F80		Nominal	µm	144	144	144	144	1	Owner	16		A	
Specific Energy		P80 9 µm	kWh/t	116.5	116.5	116.5	116.5	1	Owner	16		A	
Type				IsaMill	IsaMill	IsaMill	IsaMill	1	Owner	16		A	
Model				M3000	M3000	M3000	M3000	1	Owner	16		A	
Number				1	1	1	1	1	Owner	16		A	
Configuration				Closed Circuit	Closed Circuit	Closed Circuit	Closed Circuit	1	Owner	16		A	
Mill Power Draw		Nominal	kW	860	860	860	860	1	Owner	16		A	
Installed Motor Power			kW	1,120	1,120	1,120	1,120	1	Owner	16		A	
UFG Mill Recycle Proportion			%	0	0	0	0	1	Owner	4		A	
UFG Circuit Product Size P80		Nominal	µm	9.0	9.0	9.0	9.0	1	Owner	16		A	
UFG Circuit Product Size P80 - Range			µm	9 - 12	9 - 12	9 - 12	9 - 12	1	Owner	16		A	
Gland Seal Water			m³/h	0.0	1.2	1.2	0.0	1	Owner	2		A	
Flushing Gland Seal Water			m³/h	0.0	5.4	5.4	0.0	1	Owner	2		A	
9.2		UFG Mill 1 Trash Screen (Brownfields)			Existing	Existing	Existing	Existing					
		Feed Stream			UFG Mill 1 Discharge Vibrating Screen	UFG Mill 1 Discharge Vibrating Screen	UFG Mill 1 Discharge Vibrating Screen	UFG Mill 1 Discharge Vibrating Screen	1	Owner	43		A
		Type			1	1	1	1	1	Owner	16		A
		Number of Screens per UFG Circuit		No.	1	1	1	1	1	Owner	16		A
		Screen Solids Feed Rate per Screen	P80 9 µm	t/h	0	61	61	0	6	Calculated			
		Screen Feed Density	P80 9 µm	% w/w	45.5	42.8	42.8	43.5	6	Calculated			
		Screen Feed Flowrate per Screen	P80 9 µm	m³/h	0	91	91	0	6	Calculated			
		Screen Size		m	1.2 x 2.4	1.2 x 2.4	1.2 x 2.4	1.2 x 2.4	1	Owner	16		A
		Screen Area per Screen	W x L	m²	2.9	2.9	2.9	2.9	6	Calculated			
		Specific Screen Flux Rate		m³/m²/h	0.0	31.6	31.6	0.0	6				

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um				
9.4	New Concentrate UFG Mill 2 & 3			Greenfields Concentrate Filter Feed Tank A & B (common balance line)	Greenfields Concentrate Filter Feed Tank A & B (common balance line)	Greenfields Concentrate Filter Feed Tank A & B (common balance line)	Greenfields Concentrate Filter Feed Tank A & B (common balance line)			M15,000, 2 off	A
	Circuit Feed Streams	UFG Mill 2/3						2	EPC		0
	Nominal Feed Rate - Fresh Feed	P ₈₀ 9 µm	t/h	30.0	30.0	30.0	30.0	1	Owner	24	A
	Maximum Feed Rate - Fresh Feed	P ₈₀ 12 µm	t/h	46.0	46.0	46.0	46.0	1	Owner	24	A
	Total Feed Solids Concentration		% w/w	47.5	47.5	47.5	47.7	5	Vendor	65	A
	Feed F ₈₀	Nominal	µm	144	144	144	144	1	Owner	16	B
	Specific Energy	P ₈₀ 9 µm	kWh/t	116.5	116.5	116.5	116.5	1	Owner	16	A
	Type			IsaMill	IsaMill	IsaMill	IsaMill	1	Owner	23	A
	Model			M15,000	M15,000	M15,000	M15,000	1	Owner	23	A
	Number			2	2	2	2	1	Owner	23, 43	A
	Configuration			Closed Circuit	Closed Circuit	Closed Circuit	Closed Circuit	1	Owner	23	A
	Mill Power Draw	Nominal	kW	3,500	3,500	3,500	3,500	1	Owner	5	A
	Installed Motor Power		kW	3,800	3,800	3,800	3,800	1	Owner	23	A
	UFG Mill Recycle Proportion		%	20	20	20	5	5	Vendor	65	B
	UFG Circuit Product Size P ₈₀	Nominal	µm	9.0	9.0	9.0	9.0	1	Owner	16	A
	UFG Circuit Product Size P ₈₀ - Range		µm	9 - 12	9 - 12	9 - 12	9 - 12	1	Owner	16	A
	Gland Seal Water		m³/h	2.0	2.0	2.0	2.0	5	Vendor	65	B
	Flushing Gland Seal Water		m³/h	5.4	5.4	5.4	5.4	5	Vendor	65	B
9.5	UFG Mill 2 & 3 Trash Screens			Greenfields UFG Mill 2 & 3 Fresh Feed Inclined Static Screen	Greenfields UFG Mill 2 & 3 Fresh Feed Inclined Static Screen	Greenfields UFG Mill 2 & 3 Fresh Feed Inclined Static Screen	Greenfields UFG Mill 2 & 3 Fresh Feed Inclined Static Screen				
	Feed Stream							1	Owner	23	A
	Type							1	Owner	74	B
	Number of Screens per UFG Circuit	No.		1	1	1	1	1	Owner	23	A
	Screen Solids Feed Rate per Screen	P ₈₀ 9 µm	t/h	37	46	46	42	6	Calculated		
	Screen Feed Density	P ₈₀ 9 µm	% w/w	59.6	59.0	59.0	59.6	6	Calculated		
	Screen Feed Flowrate per Screen	P ₈₀ 9 µm	m³/h	33	41	41	37	6	Calculated		
	Screen Size	L x W	m	1,240 x 1.6	1,240 x 1.6	1,240 x 1.6	1,240 x 1.6	5	Vendor	77	B
	Screen Area per Screen		m²	2.84	2.84	2.84	2.84	5	Vendor	77	B
	Specific Screen Flux Rate		m³/m²/h	14.8	17.7	17.7	16.1	6	Calculated		
	Screen Aperture		mm	1.0	1.0	1.0	1.0	1	Owner	16	A
	Screen Material			Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	5	Vendor	77	B
	Screen Spray Water		m³/h	9.0	9.0	9.0	9.0	5	Vendor	77	B
9.6	UFG Mill 2 & 3 Classification			Greenfields 100CVX	Greenfields 100CVX	Greenfields 100CVX	Greenfields 100CVX				
	Cyclone Model			1	1	1	1	1	Owner	16	A
	Number of Cyclone Clusters per UFG Circuit			1	1	1	1	5	Vendor	54	A
	Number Installed Cyclones per Cluster	No.		24	24	24	24	5	Vendor	54	B
	Number of Cyclone Distributor Outlets Available	No.		24	24	24	24	5	Vendor	54	A
	Spare Outlets Available	No.		0	0	0	0	6	Calculated		
	Number of Operating Cyclones	No.		14	14	14	19	5	Vendor	54	B
	Number of Standby Cyclones	No.		10	10	10	5	6	Calculated		
	Cyclone Solids Feed Rate - per UFG circuit	P ₈₀ 9 µm	t/h	147.3	161.0	161.0	146.1	6	Calculated		
	Cyclone Feed Flow - per UFG circuit	P ₈₀ 9 µm	m³/h	319.0	348.5	348.5	315.1	6	Calculated		
	Cyclone Feed Solids Concentration		% w/w	32.8	32.8	32.8	32.9	5	Vendor	65	B
	Underflow Solids Concentration	P ₈₀ 9 µm	t/h	48.0	48.0	48.0	48.0	5	Vendor	65	B
	Underflow Solids - per UFG circuit	P ₈₀ 9 µm	t/h	109.9	115.0	115.0	104.4	6	Calculated		
	Overflow Solids Concentration	P ₈₀ 9 µm	% w/w	17.0	18.3	18.3	18.4	6	Calculated		
	Overflow Solids - per UFG circuit	P ₈₀ 9 µm	t/h	37.5	46.0	46.0	41.8	6	Calculated		
	Cyclone Circulating Load		%	293	250	250	250	5	Vendor	54	0
	Operating Pressure		kPa	247	247	247	248	5	Vendor	54	B
	Product Size P ₈₀		µm	9	9	9	9	1	Owner	5	A
9.7	Concentrate Pre-Leach Thickener (Brownfields)			Existing	Existing	Existing	Existing			Existing Pre-Leach Thickener 30-T1-31 Repurposed	
	Feed Streams			UFG Circuit 1, 2 & 3 Cyclone OI's	UFG Circuit 1, 2 & 3 Cyclone OI's	UFG Circuit 1, 2 & 3 Cyclone OI's	UFG Circuit 1, 2 & 3 Cyclone OI's	1	Owner	24	A
	Solids Feed Rate		t/h	75	104	104	84	6	Calculated		
	Feed Flowrate - Excl. Diluted Flocculant		m³/h	362	475	475	368	6	Calculated		
	Feed Solids Concentration		% w/w	17	18	18	18	6	Calculated		
	Underflow Solids Concentration		% w/w	55	55	55	55	1	Owner	115	1
	Calculated Flux Rate		t/m²/h	0.13	0.18	0.18	0.15	6	Calculated		
	Testwork Flux Rate		t/m²/h	0.25	0.25	0.25	0.25	1	Owner	103	A
	Thickener Area Calculated		m²	573	573	573	573	6	Calculated		
	Diameter	Existing	m	27	27	27	27	1	Owner	103	A
	Thickener Spray Water		m³/h	15	15	15	15	3	Assumed		C
10.0	CIL - FLOTATION CONCENTRATE										
10.1	General Circuit (Brownfields)			Existing CIL 2 & 3	Existing CIL 2 & 3	Existing CIL 2 & 3	Existing CIL 2 & 3				
	Solids Feed Rate	Fresh	t/h	75	104	104.1	84	6	Calculated		
	Feed Pulp Density	Range	% w/w	50 - 60	50 - 60	50 - 60	50 - 60	1	Owner	108	0
	Actual Feed Pulp Density	Total	% w/w	49	49	48.7	49	6	Calculated		
	Slurry Feed Flow Rate	Fresh	m³/h	77	107	107	86	6	Calculated		
	Slurry Feed Flow Rate	Total	m³/h	99	137	137	110	6	Calculated		
	Leach Gold Extraction		%	90	90	90	90	9	Agreed		
	Adsorption Efficiency		%	99	99	99	99	1	Owner	44	B
10.2	Leaching (Brownfields)			Existing CIL 2 Concentrate Pre-Leach Thickener Underflow & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Existing CIL 2 Concentrate Pre-Leach Thickener Underflow & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Existing CIL 2 Concentrate Pre-Leach Thickener Underflow & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Existing CIL 2 Concentrate Pre-Leach Thickener Underflow & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)			CIL 2 repurposed for leach duty	
	Feed Streams							1	Owner	106	0
	Leach pH										
	30-TK-32A - 30-TK-32C			11.75 - 12.5	11.75 - 12.5	11.75 - 12.5	11.75 - 12.5	1	Owner	108	0
	30-TK-32D			11.5 - 12.5	11.5 - 12.5	11.5 - 12.5	11.5 - 12.5	1	Owner	108	0
	30-TK-32E			11 - 12.5	11 - 12.5	11 - 12.5	11 - 12.5	1	Owner	108	0
	Number of Tanks			7	7	7	7	1	Owner	114	1
	Effective Tank Capacity	Per Tank	m³	1,685	1,685	1,685	1,685	1	Owner	16	A
	Total Tank Capacity		m³	11,795	11,795	11,795	11,795	6	Calculated		
	Residence Time		h	119	86	107	107	6	Calculated		
	Minimum Residence Time		h	80	80	80	80	1	Owner	103	A
	Oxygen Addition Points			All Tanks	All Tanks	All Tanks	All Tanks	1	Owner	16	A
	Oxygen Addition Method			Side Lances on Tanks A to F and Oxygen Circulating Pump on Tanks A and B	Side Lances on Tanks A to F and Oxygen Circulating Pump on Tanks A and B	Side Lances on Tanks A to F and Oxygen Circulating Pump on Tanks A and B	Side Lances on Tanks A to F and Oxygen Circulating Pump on Tanks A and B	1	Owner	43	4
	Dissolved Oxygen Concentration	Target	ppm	10	10	10	10	1	Owner	16	A
	Cyanide Addition Points			All Tanks	All Tanks	All Tanks	All Tanks	1	Owner	43	A
	Cyanide Concentrations										
	30-TK-32A	Target	ppm	500	500	500	500	1	Owner	108	0
	30-TK-32B	Target	ppm	1,000	1,000	1,000	1,000	1	Owner	108	0
	30-TK-32C	Target	ppm	2,200	2,200	2,200	2,200	1	Owner	108	0
	30-TK-32D	Target	ppm	5,000	5,000	5,000	5,000	1	Owner	108	0
	30-TK-32E	Target	ppm	6,500	6,500	6,500	6,500	1	Owner	108	0
	Milk of Lime Addition Points			Leach Feed Box	Leach Feed Box	Leach Feed Box	Leach Feed Box	1	Owner	43	A
	Lead Nitrate Addition Points			Leach Feed Box	Leach Feed Box	Leach Feed Box	Leach Feed Box	1	Owner	43	A
10.3	Adsorption (Brownfields)			Existing CIL 3	Existing CIL 3	Existing CIL 3	Existing CIL 3			CIL 3 repurposed for adsorption duty	
	Number of Tanks			6	6	6	6	1	Owner	114	1
	Effective Tank Capacity	Per Tank	m³	1,685	1,685	1,685	1,685	1	Owner	16	A
	Total Adsorption Tank Live Volume		m³	10,110	10,110	10,110	10,110	6	Calculated		
	Residence Time		h	102	74	92	92	6	Calculated		
	Minimum Residence Time		h	60	60	60	60	1	Owner	103	A
	Total Leach & Adsorption Time		h	221	160	160	199	6	Calculated		
	Carbon Concentration	per tank	g/L	20	20	20	20	1	Owner	16	A
	Maximum Carbon Residence Time		days	14	14	14	14	1	Owner	103	A
	Carbon Movements per Day	Max	t/d	33	33	33	33	6	Calculated		
	Carbon In Circuit	Total	t	202.2	202.2	202.2	202.2	6	Calculated		
	Design Carbon Loading		g/t	4,490	4,490	4,490	4,490	1	Owner	16, 103	A
	Carbon Transfer Method			Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	1	Owner	16	A
	Carbon Transfer Period	per cycle	h	4.0	4.0	4.0	4.0	6	Calculated		
	Carbon and Pulp Transfer Rate		m³/h	138	138	138	138	2	EPC	119	4

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
				1	2	3	4				
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um				
10.4	Loaded Carbon Recovery Screen & Transfer			Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	2	EPC		
	Loaded Carbon Advance Method			275	275	275	275	2	EPC		
	Loaded Carbon Pump & Pulp Forwarding Rate			2	2	2	2	2	EPC		
	Loaded Carbon Transfer Time	Pump to Screen									
	Feed Streams			Loaded Carbon Slurry from CIL3 Tank A (or Tank B)	Loaded Carbon Slurry from CIL3 Tank A (or Tank B)	Loaded Carbon Slurry from CIL3 Tank A (or Tank B)	Loaded Carbon Slurry from CIL3 Tank A (or Tank B)	1	Owner	As per P3-EPC-COR-602	
	Type			Linear Motion	Linear Motion	Linear Motion	Linear Motion	1	Owner	New Derrick Linear Motion Screen as per P3-EPC-COR-602	
	Number	No.		1	1	1	1	1	Owner	New Derrick Linear Motion Screen as per P3-EPC-COR-602	
	Feed Rate	m³/h		275	275	275	275	6	Calculated		
	Area	m²		3.80	3.80	3.80	3.80	5	Vendor	117	
	Deck Type			Derrick Polyweb Urethane	Derrick Polyweb Urethane	Derrick Polyweb Urethane	Derrick Polyweb Urethane	5	Vendor	117	
	Aperture	mm		0.8 x 12.8	0.8 x 12.8	0.8 x 12.8	0.8 x 12.8	5	Vendor	117	
	Total Spray Water Addition	m³/h		10	10	10	10	3	Assumed		
	Loaded Carbon Moisture	% w/w		40	40	40	40	1	Owner		
	Loaded Carbon Transfer Moisture	Maximum % w/w		55	55	55	55	1	Owner	4	Screen Oversize From Transfer Hopper
	Loaded Carbon Transfer Hopper Capacity	Minimum m³		23	23	23	23	2	EPC		
	Loaded Carbon Transfer Time	Hopper to Column	min	60	60	60	60	2	EPC		
10.5	Interstage Carbon Screens (Brownfields)			Existing CIL 3	Existing CIL 3	Existing CIL 3	Existing CIL 3				
	Type			Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	1	Owner	16	
	Supplier & Model			Kemix MPSP 1200	Kemix MPSP 1200	Kemix MPSP 1200	Kemix MPSP 1200	1	Owner	16	
	Number	per tank	No.	1	1	1	1	1	Owner	16	
	Screen Feed Rate - per Screen	w/o carbon advance	m³/h	102	140	140	113	6	Calculated		
		w/ carbon advance	m³/h	240	278	278	251	6	Calculated		
	Area Selected	Selected	m²	12.0	12.0	12.0	12.0	1	Owner	16	
	Specific Screen Flux Rate	w/ carbon advance	m³/m²/h	20.0	23.1	23.2	20.9	6	Calculated		
		Maximum	m³/m²/h	100	100	100	100	1	Owner	16	
	Screen Material			Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	1	Owner	102	
	Screen Aperture	mm		1.0	1.0	1.0	1.0	1	Owner	16	
	CIL 3 Tails Hopper										
	Feed Flowrate		m³/h	106	144	144	118	6	Calculated		
	Discharge Destination			CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	1	Owner	43	
	Cyanide Concentration	ppm		750	750	750	750	1	Owner	108	
11.0	FLOTATION TAILINGS THICKENING										
11.1	Flotation Tailings Thickener			Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	1	Owner	16	Owner Supplied
	Feed Stream										
	Solids Feed Rate		th	3,071	2,858	3,042	3,411	6	Calculated		
	Feed Flowrate - Excl. Diluted Flocculant		m³/h	6,871	6,492	6,894	6,187	6	Calculated		
	Feed Solids Concentration - Excl. Diluted Flocculant		% w/w	33.0	32.6	32.7	38.9	6	Calculated		
	Underflow Solids Concentration	Nominal	% w/w	55.9	56.0	56.0	55.9	6	Calculated		Achieve 55 % solids concentration in flotation tails CIL circuit post dilution (i.e. total feed % solids)
	Underflow Solids Concentration Range		% w/w	50-62	50-62	50-62	50-62	1	Owner	23	
	Testwork Flux Rate		t/m²/h	1.50	1.50	1.50	1.50	1	Owner	23	
	Operating Flux Rate Range		t/m²/h	1.05 - 1.27	1.05 - 1.27	1.05 - 1.27	1.05 - 1.27	1	Owner	16	
	Nominal Flux Rate		t/m²/h	1.09	1.01	1.08	1.21	6	Calculated		
	Thickener Area Calculated		m²	2,827	2,827	2,827	2,827	6	Calculated		
	Diameter Selected		m	60	60	60	60	1	Owner	16	
	Thickener Underflow Destination(s)			CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	1	Owner	43	
	Number of Underflow Discharge Pumps			2	2	2	2	1	Owner	16	
	Pump Configuration			Duty/Standby	Duty/Standby	Duty/Standby	Duty/Standby	1	Owner	43	
	Total Thickener Underflow		m³/h	3,329	3,087	3,289	3,701	6	Calculated		
12.0	CIL - FLOTATION TAILINGS										
12.1	General Circuit			CIL 4	CIL 4	CIL 4	CIL 4				
	Feed Streams			Flotation Tailings Thickener U/F, CIL 3 Tailings & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Flotation Tailings Thickener U/F, CIL 3 Tailings & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Flotation Tailings Thickener U/F, CIL 3 Tailings & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Flotation Tailings Thickener U/F, CIL 3 Tailings & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	1	Owner	106	
	Number of Trains			1	1	1	1	1	Owner	43	
	Fresh Solids Feed Flow Rate		th	3,151	2,969	3,153	3,500	6	Calculated		Flotation Tailings Thickener U/F & CIL 3 Tailings
	Total Solids Feed Flow Rate		th	3,164	2,982	3,166	3,515	6	Calculated		
	Fresh Slurry Feed Flow Rate		m³/h	3,438	3,234	3,437	3,822	6	Calculated		Flotation Tailings Thickener U/F & CIL 3 Tailings
	Total Slurry Feed Flow Rate		m³/h	3,511	3,305	3,510	3,900	6	Calculated		
	Target Feed Pulp Density	Target	% w/w	55	55	55	55	1	Owner	16, 103	Post dilution (i.e. total feed % solids)
	Maximum Feed Pulp Density		% w/w	60	60	60	60	1	Owner	43	Post dilution (i.e. total feed % solids)
	Feed Pulp Density	Actual	% w/w	55	55	55	55	6	Calculated		
	Leach Gold Extraction		%	38.0	38.0	38.0	38.0	1	Owner	44	
	Adsorption Efficiency		%	88.0	88.0	88.0	88.0	1	Owner	44	Range 84 - 92%
	Leaching			CIL 4	CIL 4	CIL 4	CIL 4				
	Leach pH			8.8 - 9.3	8.8 - 9.3	8.8 - 9.3	8.8 - 9.3	1	Owner	108	
	F21-TK-411			2	2	2	2	1	Owner	43	
	Number of Tanks	per train		2	2	2	2	1	Owner	43	
Live Tank Capacity		m³	5,600	5,600	5,600	5,600	1	Owner	43		
Total Tank Capacity		m³	11,200	11,200	11,200	11,200	6	Calculated			
Residence Time		h	3.2	3.4	3.2	2.9	6	Calculated			
Oxygen Addition Points			All Leach Tanks	All Leach Tanks	All Leach Tanks	All Leach Tanks	1	Owner	43		
Oxygen Addition Method			Side Lances and Circulating Pumps on both Leach Tanks	Side Lances and Circulating Pumps on both Leach Tanks	Side Lances and Circulating Pumps on both Leach Tanks	Side Lances and Circulating Pumps on both Leach Tanks	1	Owner	43		
Dissolved Oxygen Concentration	Target	ppm	15	15	15	15	1	Owner	108		
Cyanide Addition Points			Feed Box and Tank 2	Feed Box and Tank 2	Feed Box and Tank 2	Feed Box and Tank 2	1	Owner	43		
Cyanide Concentration		ppm	175	175	175	175	1	Owner	108		
F21-TK-411											
Milk of Lime Addition Points			Leach Feed Box	Leach Feed Box	Leach Feed Box	Leach Feed Box	1	Owner	43		
12.3	Adsorption			CIL 4	CIL 4	CIL 4	CIL 4				
	Number of Tanks	per train		6	6	6	6	1	Owner	43	Provision in the layout for a future, 7th adsorption tank
	Live Tank Capacity		m³	5,600	5,600	5,600	5,600	1	Owner	43	
	Total Adsorption Tank Live Volume		m³	33,600	33,600	33,600	33,600	6	Calculated		
	Residence Time		h	9.6	10	10	9	6	Calculated		
	Total Leach & Adsorption Time		h	12.8	13.6	12.8	11.5	6	Calculated		2 Leaching & 6 Adsorption Tanks will not achieve the 14 hours total residence time. Client is aware and has advised to maintain 8 tanks with the 9th as a future.
	Oxygen Addition Points			First Two Adsorption Tanks	First Two Adsorption Tanks	First Two Adsorption Tanks	First Two Adsorption Tanks	1	Owner	43	
	Oxygen Addition Method			Side Lances	Side Lances	Side Lances	Side Lances	1	Owner	43	
	Carbon Concentration - Duty	per tank	g/L	12.5	12.5	12.5	12.5	1	Owner	109	
	Carbon Concentration Used for Carbon Transfer		g/L	12.5	12.5	12.5	12.5	1	Owner	109	
	Carbon Movements per Day	Max	t/d	30	30	30	30	6	Calculated		
	Carbon Residence Time	Maximum	d	14	14	14	14	1	Owner	103	
	Carbon In Circuit	@ 8 g/L	t	269	269	269	269	6	Calculated		
		@ 10 g/L	t	336	336	336	336	6	Calculated		
		@ 12.5 g/L	t	420	420	420	420	6	Calculated		
	Nominal Carbon Loading		g/t	461	461	461	461	1	Owner	16, 103	
Carbon Loading Range		g/t	332 - 579	332 - 579	332 - 579	332 - 579	1	Owner	16, 103		
Carbon Transfer Method			Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	2	EPC			
Carbon Transfer Period		h	5.5	5.5	5.5	5.5	6	Calculated			
Carbon and Pulp Transfer Rate		m³/h	220	220	220	220	2	EPC			
Loaded Carbon Advance Method			Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	2	EPC			
Loaded Carbon Pump & Pulp Forwarding Rate		m³/h	280	280	280	280	2	EPC			
Loaded Carbon Transfer Time		h	4.3	4.3	4.3	4.3	6	Calculated			
12.4	Loaded Carbon Recovery Screen			CIL 4	CIL 4	CIL 4	CIL 4				
	Feed Streams			First Adsorption Tank	First Adsorption Tank	First Adsorption Tank	First Adsorption Tank	1	Owner	16	
	Type			Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	1	Owner	16	
	Number	No.		1	1	1	1	1	Owner	16	

[illegible]

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
14.3	Acid Washing										
	Acid Washes Per Week	No. t		12			6	Calculated			
	Acid Wash Column Capacity			15			1	Owner	8	Combined Acid Wash & Elution column	A
	Acid Type			Hydrochloric Acid			1	Owner	103		A
	Acid Dosing Method			Pumped			1	Owner	103		A
	Acid Concentration	% w/v HCl		3			1	Owner	103		A
	Acid Injection Time	min		17			1	Owner	103		A
	Acid Soak Time	min		60			1	Owner	103		A
	Acid Water Rinse Time	min		120			1	Owner	103		A
	Total Acid Wash Solution Flowrate	BV/h		2			1	Owner	8		A
	Total Acid Wash Solution Flowrate	m³/h		63.8			6	Calculated			
	Dilute Acid Flowrate	BV		0.6			6	Calculated			
	Dilute Acid Solution Used	m³		63.8			6	Calculated			
	Neat Acid Solution Used	m³		18.085			6	Calculated			
	Waste Acid Solution Destination			1.5			6	Calculated			
	Acid Wash Temperature	°C		Final Tails Tank Ambient			1	Owner	43		B
							1	Owner	44		A
14.4	Water Rinse										
	Water Flowrate	m³/h		63.8			6	Calculated		Based on 2 BV/h	
	Acid Water Rinse Time	min		120			1	Owner	103		A
	Volume Water Used	m³		128			6	Calculated			
	No. Bed Volumes	BV		4			1	Owner	8, 103		
	Solution Destination			Final Tails Tank Ambient			1	Owner	43		B
	Acid Rinse Temperature	°C		Scheme Water			1	Owner	44		A
	Rinse Water Source						1	Owner	103		A
14.5	Elution										
	Elution Type			Split AARL			1	Owner	43		A
	Required Elutions Per Week	No. t		12			1	Owner	8, 43		B
	Elution Column Capacity			15			1	Owner	8, 43		A
	Carbon Bed Volume	m³		31.9			6	Calculated			
14.6	Elution Column Operating Pressure	Top of Column kPa(g)		340			2	EPC		To prevent flashing (boiling) of solution inside column.	B
	Pre-treatment (with Lean Eluate)										
	Cycle Time	min		30			2	EPC			A
	Solution Flowrate	BV/h		2			1	Owner	8		A
	Temperature	°C		63.8			6	Calculated			
	No. of Bed Volumes			95			2	EPC			A
	Pre-soak Solution Volume	m³		1.0			6	Calculated		5 BVs of Pregnant Eluate required by owner (Source 8)	
	Column Discharge Destination			31.9			6	Calculated			
	Solution Strength	% w/v NaCN		Lean Eluate Tank			8	Industry			B
				1.7			1	Owner	103		A
	Sodium Cyanide Consumption	kg/pre-treat		2			1	Owner	103		A
	Caustic Consumption	m³/pre-treat	Neat Solution	543			6	Calculated			
	Lean Eluate Solution Make-up Per Mix	m³/pre-treat	Neat Solution	1.51			6	Calculated			
		m³/mix		638			6	Calculated			
14.7	Elution (with Lean Eluate)										
	Elution Cycle Time	min		0.85			6	Calculated			
	Elution Solution Flowrate	BV/h		29.5			6	Calculated			
	Eluate Consumed	BV									
		m³		150			2	EPC			A
	Elution Temperature	°C		2			1	Owner	8		A
	Column Discharge Destination			63.8			6	Calculated			
				5.0			6	Calculated		5 BVs of Pregnant Eluate required by owner (Source 15)	
				159.6			6	Calculated			
				130			2	EPC			A
14.8	Elution & Cooling (with Scheme Water)			CIL4 Eluate Circulation Tanks			8	Industry			A
	Cycle Time - Scheme Water (with heaters)	min									
	Elution Temperature	°C		90			6	Calculated			
	Water Flowrate	BV/h		130			2	EPC			C
	Water Consumed	m³		2			1	Owner	8		A
				63.8			6	Calculated			
				3.0			6	Calculated			
				95.7			6	Calculated			
	Cycle Time - Scheme Water (no heaters, cooling)	min		60			2	EPC			A
	Elution Cookdown Temperature	°C		-80			2	EPC			A
	Water Flowrate	BV/h		2			1	Owner	8		A
	Water Consumed	m³		63.8			6	Calculated			
				2.0			6	Calculated			
				63.8			6	Calculated			
	Total Cycle Time	min		150			2	EPC			A
	Water Flowrate	BV/h		2			1	Owner	8		A
	Total Water Consumed	m³		63.8			6	Calculated			
				5.0			6	Calculated			
				159.6			6	Calculated		Total of 5 BVs of Lean Eluate required by owner (Source 8)	
	Column Discharge Destination			CIL4 Eluate Circulation Tanks and Lean Eluate Tank			8	Industry		When the CIL4 eluate circulation tank is full, discharge switches to the lean eluate tank	A
	Carbon Transfer In	min		30.0			6	Calculated			
	Total Acid Time (including rinsing)	h		0.5			6	Calculated			
		min		197			6	Calculated			
		h		3.3			6	Calculated			
	Total Elution Time (including Cooling)	min		330			6	Calculated			
		h		5.5			6	Calculated			
	Carbon Transfer Out	min		53.5			6	Calculated			
		h		0.9			6	Calculated			
	Total	h		10.2			6	Calculated			
14.9	Heating										
	Elution Heater Type			LPG Fired Elution Heater			1	Owner	43		A
14.10	Lean Eluate Tank										
	Total Volume of Elution Eluate Per Strip	Pregnant m³		160			6	Calculated			
	Number of Lean Eluate Tanks			1			1	Owner	43		A
14.11	Elution Water Tank										
	Total Live Volume Elution Water Tank	Minimum m³		160			6	Calculated			
14.12	Eluate Expansion Tank										
	Number of Eluate Expansion Tanks			1			2	EPC			B
14.13	Total Water Consumption Per Strip										
	Carbon Transfer	Scheme m³/strip		3.5			6	Calculated			
14.14	Elution Electrowinning										
	Carbon Wash	Scheme m³/strip		8			6	Calculated			
	Water Rinse	Scheme m³/strip		17			6	Calculated			
	Water Elution & Cooling	Scheme m³/strip		128			6	Calculated			
	Carbon Eduction (Elution to Regen)	Scheme m³/strip		160			6	Calculated			
	Carbon Quench to Sizing Screen	Scheme m³/strip		60			6	Calculated			
	Total Per Strip	Scheme m³/strip		60			6	Calculated			
				432			6	Calculated			
	Number Circuits			1			2	EPC			A
	Number of Cells Per Circuit			2			5	Vendor	48	2 off Split (2 in 1) Clean-in-Place cells	A
	Cell Configuration			Parallel - 2 off Split (2 in 1)			5	Vendor	48		B
	Cell Size	m x m		1.00 x 1.00			5	Vendor	48		A
	Effective Cross Sectional Area	m²		1.00			6	Calculated			
	Pregnant Eluate Volume	m³		160			6	Calculated			
	Pregnant Eluate Grade	g Au/m³		52.6			6	Calculated			
15.0	Goldroom Operation (Brownfields)										
	Feed Sources			Existing							
	Sludge Filter Operating Days	days/week		Flotation Con EW product (CIL2/3) & Flotation Tails EW product (CIL 4)			1	Owner	43		A
	Sludge Dewatering Cone Capacity	m³		2			1	Owner	16		A
	Sludge Filter Make			1.3			1	Owner	16		A
	Sludge Filter Model			TEFSA			1	Owner	16	470x470mm plate filter, 6 bar design pressure	A
				HPLE			1	Owner	16		
	Mercury Retort Oven Operating Days	days/week		2			1	Owner	16		A
	Mercury Retort Oven Make			Locchead			1	Owner	16		A
	Mercury Retort Oven Model			F66OV66			1	Owner	16	20 ft³ electric powered mercury retort	A
	Calcine per week	kg		800 - 1,200			1	Owner	16	Existing, Calcine logs, Max Capacity 1,600 kg per cycle	A
	Smelting Furnace Operating Days	days/week		2			1	Owner	16		A
	Furnace Make			Furnace Engineering			1	Owner	16		A
	Furnace Model			Westofen KBU 80/H			1	Owner	16	2000 L x 3000 D x 1710 H, Weishaupt G3-1a burner	A
15.1	Furnace Extraction Fan			75 kW / 1,480 RPM			1	Owner	16	IP55, 415V with 315xSSPB motor pulley and 212xSSPB fan pulley	A
	Furnace Scrubber			Metalcraft SN50247/1			1	Owner	16		A
	Smelts per week			6			1	Owner	16	Two downloads per week, three pours per day	A

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV		
				1	2	3	4						
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um						
16.0 CARBON REGENERATION - FLOTATION CONCENTRATE													
16.1	Barren Carbon Transfer (Brownfields)			Existing									
	Transfer Water Source		Carbon Transfer Water (Scheme Water)				1	Owner	16, 103		A		
	Carbon Transfer Time	min	30				1	Owner	16, 103		A		
	Carbon Transfer Concentration	% w/w	20				1	Owner	57		B		
	Transfer Water Flowrate	Each m³/h	88				6	Calculated		Based off 11t elution column			
	Transfer Water Required	Each m³	44				6	Calculated		Based off 11t elution column			
16.2	Regeneration Kiln Dewatering Screens (Brownfields)			Existing									
	Type		Horizontal Vibrating Screen				1	Owner	16, 103		A		
	Number	No.	3				1	Owner	16, 103	1x Per Regeneration Kiln	A		
	Specific Screen Flux Rate	Max Recommended t/m²/h	4				1	Owner	16, 103		A		
	Selected Size	Width m	1.2				1	Owner	16, 103		A		
		Length m	2.4				1	Owner	16, 103		A		
	Area	m²	2.9				6	Calculated					
	Feed % Solids	% w/w	20				1	Owner	57		B		
	% Mass Split to Oversize	%	100				2	EPC			B		
	Oversize Moisture %	% w/w	60				1	Owner	57		B		
16.3	Carbon Regeneration Kiln (Brownfields)			Existing									
	Regeneration Kiln Type		Horizontal				1	Owner	16, 103		A		
	Regeneration Kiln Model		Metso RK600/650HR				1	Owner	16, 103		A		
	Number		3				1	Owner	16, 103		A		
	Eluted Carbon to be Regenerated	%	100				1	Owner	16, 103		A		
	Regenerated Carbon per Batch	t	11				6	Calculated					
	Temperature	°C	750				1	Owner	16, 103		A		
	Required Kiln Operating time	h/batch	20				1	Owner	100		A		
	Capacity - Nominal	kg/h	550				6	Calculated					
	Capacity - Design	kg/h	650				1	Owner	100		A		
	Carbon Quench Tank (Brownfields)			Existing									
	Carbon Quench Tank Volume	Each m³	12				1	Owner	16, 103	3 off Quench Tanks	A		
	Carbon Transfer Time	Each min	30				1	Owner	16, 103		A		
Carbon Transfer Concentration	% w/w	20				1	Owner	57		B			
Transfer Water Flowrate	Each m³/h	88				6	Calculated		Based off 11t elution column				
Transfer Water		Carbon Transfer Water (Scheme Water)				1	Owner	16, 103		B			
	Transfer Water Required	Each m³	44				6	Calculated					
16.5	CIL 3 Barren Carbon Sizing Screen			New									
	Type		Horizontal Vibrating Screen				1	Owner	100		A		
	Number	No.	1				1	Owner	100		A		
	Feed Rate	m³/h	98				6	Calculated					
	Feed Rate	t/h	22				6	Calculated		Based off 1 x 11t elution columns			
	Selected Size	Width m	1.53				5	Vendor	84	Based off 1 x 11t elution columns	0		
		Length m	3.66				5	Vendor	84		0		
	Area	m²	5.60				6	Calculated					
	Specific Screen Flux Rate	Actual m³/m²/h	17.5				6	Calculated		w/o screen spray water			
	Deck Type		Polyurethane				1	Owner	100		A		
	Aperture	mm	1.25 x 18.5				5	Vendor	84		1		
	% Mass Split to Oversize	%	100				2	EPC	84		B		
	Oversize Moisture %	% w/w	40				1	Owner	4		B		
	Screen Spray Water	m³/h	13				5	Vendor			0		
	16.6	Regeneration Kiln Off-Gas Treatment (Brownfields)			Existing								
		Number		1				1	Owner	16, 103		A	
		Design Dust Load	mg/Nm³	375				1	Owner	16, 103		A	
Design Mercury		g/h	546				1	Owner	16, 103		A		
Design VOC		g/h	831				1	Owner	16, 103		A		
Design Gas Temperature		°C	250				1	Owner	16, 103		A		
Water Type			Process				1	Owner	16, 103		A		
Packed Bed Wet Scrubber			Existing										
Number			1				1	Owner	16, 103		A		
Design Gas Flowrate		Nm³/h	6,481				1	Owner	16, 103		A		
Design Gas Temperature		°C	80				1	Owner	16, 103		A		
Water Type			Process				1	Owner	16, 103		A		
Mist Eliminator			Existing										
Number			1				1	Owner	16, 103		A		
Design Gas Flowrate		Nm³/h	3,916				1	Owner	16, 103		A		
Design Gas Temperature		°C	58				1	Owner	16, 103		A		
Water Type			Scheme				1	Owner	16, 103		A		
Regenerative Thermal Oxidiser			Existing										
Number			1				1	Owner	16, 103		A		
Design Gas Flowrate		Nm³/h	4,132				1	Owner	16, 103		A		
Design Gas Temperature		°C	58				1	Owner	16, 103		A		
Carbon Filters			Existing										
Number			2 (Duty / Standby)				1	Owner	16, 103		A		
Design Gas Flowrate		Nm³/h	4,503				1	Owner	16, 103		A		
Design Gas Temperature		°C	80				1	Owner	16, 103		A		
Design Mercury Loading		g/h	387				1	Owner	16, 103		A		
Stack Emission Limits													
Mercury		ppmv	0.20				1	Owner	16, 103	FIM ERP Commissioning Report	A		
VOC		ppmv	3.2				1	Owner	16, 103		A		
17.0 CARBON REGENERATION - FLOTATION TAILINGS													
17.1		Barren Carbon Transfer			New - Flotation Tailings								
		Transfer Water Source		Carbon Transfer Water (Scheme Water)				1	Owner	103		B	
		Carbon Transfer Time	min	53				2	EPC			0	
		Carbon Transfer Concentration	%w/w	20				1	Owner	57		B	
		Transfer Water Flowrate	m³/h	67				6	Calculated		Based off 1x 15t carbon batch		
		Transfer Water Required	m³	60				6	Calculated		Based off 1x 15t carbon batch		
17.2		Regeneration Kiln Dewatering Screen			New								
	Type		Horizontal Vibrating Screen				5	Vendor	84		A		
	Number	No.	1				1	Owner	103	One elution column	A		
	Feed Flow Rate	m³/h	75				6	Calculated					
	Solids Feed Rate	t/h	17				6	Calculated					
	Selected Size	Width m	1.53				5	Vendor	84		0		
		Length m	3.66				5	Vendor	84		0		
	Area	m²	5.6				6	Calculated					
	Specific Screen Flux Rate	Maximum t/m²/h	4				1	Owner	103		B		
		Actual m³/m²/h	13.4				6	Calculated					
	Deck Type		Polyurethane				6	Calculated					
	Aperture	mm	1.2 x 18.5				5	Vendor	84		B		
	Feed % Solids	% w/w	20				5	Vendor	84		0		
	% Mass Split to Oversize	%	100				1	Owner	57		B		
	Oversize Moisture %	Nominal Maximum	40				2	EPC	52		B		
			60				1	Owner	57		B		
	Screen Spray Water	m³/h	13				5	Vendor			0		
	17.3	Carbon Regeneration Kiln			New								
		Reactivation Kiln Type		Horizontal - Indirect Fired, Electric				1	Owner	103		B	
		Number		1				1	Owner	103		A	
		Regenerated Carbon per Batch	t	30				6	Calculated		2 strips per day from 15t elution column		
		Regen Kiln Feed Hopper Capacity	% of full column size	200				1	Owner	103		B	
		Regen Kiln Feed Hopper Capacity	t	30				6	Calculated				
Required Kiln Operating time		Nominal h/batch	20				8	Industry			A		
Capacity - Nominal		kg/h	1,600				1	Owner	43		A		
Operating Temperature		°C	650 - 750				1	Owner	102		A		
Design Operating Temperature		°C	750				1	Owner	103				
Fraction of Carbon Regenerated		%	100				1	Owner	103				
Regenerated Carbon Mercury Grade		Nominal Range	1.3				1	Owner	16		B		
			0 - 4				1	Owner	16		B		
17.4		Carbon Quench Tank			New								
	Carbon Quench Tank Volume	Minimum m³	20				1	Owner			A		
	Carbon Transfer Time	min	53				2	EPC			0		
	Carbon Transfer Concentration	% w/w	20				2	EPC			C		
	Transfer Water Flowrate	m³/h	67.3				6	Calculated		Based off one 15t elution column			
	Transfer Water		Quench Hopper filled with Scheme Water Carbon Transfer completed with Raw Water				1	Owner	100		0		
	Transfer Water Required	m³	60				6	Calculated		Based off one 15t elution column			
	CIL 4 Carbon Sizing Screen			New									
Type		Horizontal Vibrating Screen				1	Owner	100		A			
Number	No.	1				1	Owner	103	Single CIL4 Train	A			
Feed Flow Rate	m³/h	75				6	Calculated						
Solids Feed Rate	t/h	17				6	Calculated						
Selected Size	Width m	1.53				5	Vendor	84		0			
	Length m	3.66				5	Vendor	84		0			
Area	m²	5.6				6	Calculated						
Specific Screen Flux Rate	Maximum m³/m²/h	35.0				1	Owner	103		B			
	Actual t/m²/h	3.0				6	Calculated						
Deck Type		Polyurethane				6	Calculated		w/o screen spray water				
Aperture	mm	1.25 x 18.5				5	Vendor	84		B			
% Mass Split to Oversize	%	100				5	Vendor	84		1			
Oversize Moisture %	% w/w	40				2	EPC	84		B			
						1	Owner	4		B			
Screen Spray Water	m³/h	13				5	Vendor			0			

AREA	PARAMETER		UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
				1	2	3	4					
				FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um				
17.6	Regeneration Kiln Off-Gas Treatment					New						
	Mercury Recovery	Target	%			>92			1	Owner	43	
	Design Mercury	New - Range	g/t			20 - 72			1	Owner	16	
		New - Design	g/h			108			6	Calculated		
	VOC	New	g/h			639			6	Calculated		
	Gas Temperature	New - Nominal	°C			316			5	Vendor	52	
	Gas Flow	New - Design	Nm³/h			400			5	Vendor	52	
		New - Nominal	Nm³/h			1,335			6	Calculated	52	
		New - Design	Nm³/h			3,003			6	Calculated	52	
	Carbon Fines Knock-Out Box											
	Number					1			5	Vendor	52	
	Dimensions		m			1.22 x 1.8			5	Vendor	52	
	Water Type	Sprays				Scheme			5	Vendor	52	
	Water Flowrate		m³/h			8.39			3	Assumed		
	Kiln Off-Gas Condenser											
	Number					1			5	Vendor	52	
	Cooler Type					Shell and Tube			5	Vendor	52	
	Water Type					Cooling Water (Scheme)			5	Vendor	52	
	Kiln Mist Eliminator											
	Number					New						
	Dimensions		m			1			5	Vendor	52	
	Water Type	Sprays				1.22 x 1.8			5	Vendor	52	
	Water Flowrate		m³/h			Scheme			5	Vendor	52	
						8.39			3	Assumed		
	Kiln Duct Heater											
	Number					1			5	Vendor	52	
	Power		kW			7.5			5	Vendor	52	
	Kiln Adsorption Vessel											
	Number					1			5	Vendor	52	
	Type					SIC Bed - CGS Dual Banjo			5	Vendor	52	
	Dimensions		m			1.98 x 2.7			5	Vendor	52	
	Carbon Type & Size		mm			3mm Mersorb HT			5	Vendor	52	
	Kiln Exhaust Fan											
	Number					1			5	Vendor	52	
	Design Duty		Am³/h			3,003			5	Vendor	52	
	Power		kW			12.5			5	Vendor	52	
	Kiln Discharge Stack											
	Number					1			1	Owner	52	
	Height		m			35			1	Owner	52	
	Stack Emission Limits											
	Mercury	Maximum	ppmv			0.20			1	Owner	43	
			micrograms/m³			1,800			1	Owner	52	
	Mercury	Vendor System	ppmv			0.027			6	Calculated		
	VOC	Maximum	ppmv			3.20			1	Owner	43	
18.0	CYANIDE DESTRUCT											
	Feed Streams	1							1	Owner	35	
		2							1	Owner	35	
		3							1	Owner	35	
	Decant Water Recovered from Tailings	% of Total Tailings Water	%			49			1	Owner	35	
	Decant Water from Kaltails		m³/h			728			1	Owner	44	
	Total Feed Flow to Cyanide Water Tank		m³/h			1,940			6	Calculated		
	Method					Hydrogen Peroxide			1	Owner	35	
	Residual WAD Cyanide Target	Stage 2				<1.5			1	Owner	35	
	Dosing Points					Inline Mixers to each Decant Water line (3 off)			1	Owner	35	
		Location				3			1	Owner	35	
		Number				3			1	Owner	35	
	Dosing Method					Duty/duty/duty/standby pumps			1	Owner	35	
	Dosing Control Method					Online CN _{WAD} Analyser			1	Owner	35	
19.0	REAGENTS											
19.1	Activator - Copper Sulphate											
	Delivery Form		L			Bulk Liquid			1	Owner	64	
	Delivery Size per Trailer					24,000			5	Vendor	112	
	Number of Trailers per delivery	Maximum				2			1	Owner	64	
	Delivery Size		L			48,000			6	Calculated		
	Delivery System					Compressed Air Displacement			1	Owner	64	
	Delivery Side					Central Rear			1	Owner	64	
	Solution Mix Strength (as CuSO ₄ .5H ₂ O)		% w/v			25			1	Owner	16	
	Solution Mix Strength (as CuSO ₄)		% w/v			16			1	Owner	64	
	Supply Solution SG		t/m³			1.17			1	Owner	16	
	Storage Volume	Minimum	# deliveries			2.5			1	Owner	83	
	Number of Storage Tanks					1			1	Owner	16	
	Live Volume per Tank		m³			120			6	Calculated		
	Residence Time	Actual	d			11			6	Calculated		
	Consumption											
	SAG Mill Discharge	Nominal	g/t fresh feed to Cyclone Feed Hopper			35			1	Owner	23, 75	
	Cleaner Scavenger	Nominal	g/t fresh cleaner scav feed			10			1	Owner	23, 75	
	SAG Mill Discharge	Design	g/t fresh feed to Cyclone Feed Hopper			70			6	Calculated		
	Cleaner Scavenger	Design	g/t fresh cleaner scav feed			20			6	Calculated		
	SAG Mill Discharge	Nominal	L/h			440			6	Calculated		
	Cleaner Scavenger	Nominal	L/h			7.3			6	Calculated		
	SAG Mill Discharge	Design	L/h			881			6	Calculated		
	Cleaner Scavenger	Design	L/h			14.7			6	Calculated		
	Nominal Consumption	as CuSO ₄ .5H ₂ O	L/h			447			6	Calculated		
		as CuSO ₄ .5H ₂ O	m³/h			0.45			6	Calculated		
	Design Consumption	as CuSO ₄ .5H ₂ O	L/h			896			6	Calculated		
		as CuSO ₄ .5H ₂ O	m³/h			0.90			6	Calculated		
19.2	Collector - PAX											
	Delivery Form		L			Bulk Liquid			1	Owner	64	
	Delivery Size per Trailer					24,000			5	Vendor	112	
	Number of Trailers per delivery	Maximum				2			1	Owner	64	
	Delivery Size		L			48,000			6	Calculated		
	Delivery System					Compressed Air Displacement			1	Owner	16	
	Delivery Side					Central Rear			1	Owner	64	
	Solution Mix Strength		% w/v			20			1	Owner	64	
	Supply Solution SG		t/m³			1.095			1	Owner	23	
	Number of Storage Tanks					1			1	Owner	16	
	Live Volume per Tank		m³			270			6	Calculated		
	Residence Time	Actual	d			7			2	EPC		
		Required	d			7			1	Owner	10, 16	
	Consumption											
	Rougher Feed	Nominal	g/t fresh mill feed			40			1	Owner	16, 75	
	Scavenger Feed	Nominal	g/t fresh mill feed			35			1	Owner	16, 75	
	Cleaner Conditioning	Nominal	g/t fresh mill feed			16			1	Owner	16, 75	
	Cleaner Scavenger	Nominal	g/t fresh mill feed			16			1	Owner	16, 75	
	Rougher Feed	Nominal	L/h			629			6	Calculated		
	Scavenger Feed	Nominal	L/h			551			6	Calculated		
	Cleaner Conditioning	Nominal	L/h			252			6	Calculated		
	Cleaner Scavenger	Nominal	L/h			252			6	Calculated		
	Total Consumption	Nominal	L/h			1,683			6	Calculated		
		Nominal	m³/h			1.68			6	Calculated		
	Consumption											
	Rougher Feed	Design	g/t fresh mill feed			80			6	Calculated		
	Scavenger Feed	Design	g/t fresh mill feed			70			6	Calculated		
	Cleaner Conditioning	Design	g/t fresh mill feed			32			6	Calculated		
	Cleaner Scavenger	Design	g/t fresh mill feed			32			6	Calculated		
	Rougher Feed	Design	L/h			1,258			6	Calculated		
	Scavenger Feed	Design	L/h			1,101			6	Calculated		
	Cleaner Conditioning	Design	L/h			503.3			6	Calculated		
	Cleaner Scavenger	Design	L/h			503.3			6	Calculated		
	Total Consumption	Design	L/h			3,366			6	Calculated		
		Design	m³/h			3.37			6	Calculated		

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
19.3	Frother - Luprofroth 168										
	Delivery Form			Bulk Liquid	Bulk Liquid	Bulk Liquid		1	Owner	64	A
	Delivery Size per Trailer			26,000	26,000	26,000		5	Vendor		0
	Number of Trailers per delivery	Maximum		1	1	1		1	Owner	16	A
	Delivery Size	L		26,000	26,000	26,000		6	Calculated		
	Delivery System			Pumped Transfer From Delivery Truck	Pumped Transfer From Delivery Truck	Pumped Transfer From Delivery Truck	Pumped Transfer From Delivery Truck	1	Owner	64	A
	Delivery Side			Passenger	Passenger	Passenger	Passenger	1	Owner	64	B
	Solution Mix Strength	% w/w		100	100	100	100	1	Owner	64	A
	Supply Solution SG	tm³		1.082	1.082	1.082	1.082	1	Owner	16	A
	Number of Storage Tanks			1	1	1	1	2	EPC		A
	Live Volume per Tank	m³		60	60	60	60	1	Owner	16	A
	Residence Time	Minimum		7	7	7	7	1	Owner	10	A
		Actual		29	30	29	26	6	Calculated		
										Residence time calculated using nominal consumption rates.	
	Consumption										
	Rougher Feed	Nominal	g/t fresh mill feed	12	12	12	12	1	Owner	16, 75	A
	Scavenger Feed	Nominal	g/t fresh mill feed	6	6	6	6	1	Owner	16, 75	A
	Cleaner Feed	Nominal	g/t fresh mill feed	6	6	6	6	1	Owner	44, 75	A
	Cleaner Scavenger	Nominal	g/t fresh mill feed	6	6	6	6	1	Owner	16, 75	A
	Rougher Feed	Nominal	L/h	34.9	32.9	34.9	38.8	6	Calculated		
	Scavenger Feed	Nominal	L/h	17.4	16.4	17.4	19.4	6	Calculated		
	Cleaner Feed	Nominal	L/h	17.4	16.4	17.4	19.4	6	Calculated		
	Cleaner Scavenger	Nominal	L/h	17.4	16.4	17.4	19.4	6	Calculated		
	Total Consumption	Nominal	L/h	87.2	82.1	87.2	96.9	6	Calculated		
			m³/h	0.09	0.08	0.09	0.10	6	Calculated		
	Consumption										
	Rougher Feed	Design	g/t fresh mill feed	24	24	24	24	6	Calculated		100% Design Factor for pumping
	Scavenger Feed	Design	g/t fresh mill feed	12	12	12	12	6	Calculated		100% Design Factor for pumping
	Cleaner Feed	Design	g/t fresh mill feed	12	12	12	12	6	Calculated		100% Design Factor for pumping
	Cleaner Scavenger	Design	g/t fresh mill feed	12	12	12	12	6	Calculated		100% Design Factor for pumping
	Rougher Feed	Design	L/h	69.8	65.7	69.8	77.5	6	Calculated		100% Design Factor for pumping
	Scavenger Feed	Design	L/h	34.9	32.9	34.9	38.8	6	Calculated		100% Design Factor for pumping
	Cleaner Feed	Design	L/h	34.9	32.9	34.9	38.8	6	Calculated		100% Design Factor for pumping
	Cleaner Scavenger	Design	L/h	34.9	32.9	34.9	38.8	6	Calculated		100% Design Factor for pumping
	Total Consumption	Design	L/h	174.4	164.3	174.4	193.8	6	Calculated		100% Design Factor for pumping
			m³/h	0.17	0.16	0.17	0.19	6	Calculated		
19.4	Flocculant										
	Packaging Type			Bulk Powder	Bulk Powder	Bulk Powder	Bulk Powder	1	Owner	64	A
	Delivery Size per Trailer			22,500	22,500	22,500	22,500	1	Owner	64	A
	Number of Trailers per delivery	Maximum		1	1	1	1	1	Owner	64	A
	Delivery Size	kg		22,500	22,500	22,500	22,500	6	Calculated		
	Delivery System			Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	1	Owner	64	C
	Delivery Side			Passenger	Passenger	Passenger	Passenger	1	Owner	64	B
	Storage Volume	Minimum	# deliveries	2	2	2	2	1	Owner	23	A
	Number of Mixing Tanks			1	1	1	1	1	Owner	16	A
	Mixing Tank Live Volume		m³	40	40	40	40	5	Vendor	36	B
	Solution Mix Strength	% w/v		0.30	0.30	0.30	0.30	1	Owner	44	A
	Mixes per Day per Tank			14	14	15	16	6	Calculated		Based on nominal Stage 2 consumption
	Flocculant Dilution Factor			10	10	10	10	1	Owner	103	B
	Number of Storage Tanks			1	1	1	1	5	Vendor	36	A
	Storage Tank Live Volume		m³	450	450	450	450	5	Vendor	36	A
	Residence Time		h	19	19	18	17	6	Calculated		Vendor advice. Based on nominal Stage 2 consumption
	Consumption										
	Flotation Tailings Thickener	Nominal	g/t	20	20	20	20	1	Owner	23	A
		Nominal	kg/h	61.4	57.2	60.8	68.2	6	Calculated		
		Nominal	L/h	20,473	20,054	22,738	22,738	6	Calculated		
		Design	L/h	40,946	38,108	40,557	45,476	6	Calculated		
	Concentrate Pre-Leach Thickener	Nominal	g/t	70	70	70	70	1	Owner	23	A
		Nominal	kg/h	5.2	7.3	7.3	5.8	6	Calculated		
		Nominal	L/h	1,748	2,430	2,430	1,948	6	Calculated		
		Design	L/h	3,496	4,859	4,860	3,897	6	Calculated		
	Concentrate Thickener	Nominal	g/t	70	70	70	70	1	Owner	23	A
		Nominal	kg/h	5.2	7.3	7.3	5.8	6	Calculated		
		Nominal	L/h	1,748	2,430	2,430	1,948	6	Calculated		
		Design	L/h	3,496	4,859	4,860	3,897	6	Calculated		
	Total Nominal Consumption	Stage 2	g/h	71,909	71,742	75,417	79,908	6	Calculated		
		Stage 2	L/h	23,970	23,914	25,139	26,636	6	Calculated		
		Stage 2	m³/h	24.0	23.9	25.1	26.6	6	Calculated		
		Stage 2	t/a	588	587	617	654	6	Calculated		
	Total Design Consumption	Stage 2	g/h	95,000	95,000	95,000	95,000	2	EPC		Based on 28.35Mtpa Total Consumption for Stage 2 when the Regrind Mill was still included. Regrind Mill was moved post tender phase.
		Stage 2	t/a	777	777	777	777	6	Calculated		
		Stage 2	m³/h	31.7	31.7	31.7	31.7	6	Calculated		Neat Concentration
		Stage 2	L/h	47,938	47,826	50,277	53,270	6	Calculated		100% Design Factor
		Stage 2	m³/h	47.9	47.8	50.3	53.3	6	Calculated		100% Design Factor
19.6	Silo Capacity	Actual	t	45	45	45	45	5	Vendor	36	A
			d	26	26	25	23	6	Calculated		Based on nominal Stage 2 consumption
	pH Modifier - Quicklime										
	Type			New Powder	New Powder	New Powder	New Powder	1	Owner	64	B
	Packaging Type			Bulk	Bulk	Bulk	Bulk	1	Owner	64	A
	Delivery Size per Trailer	kg		30,000	30,000	30,000	30,000	1	Owner	64	A
	Number of Trailers Per Delivery	#		3	3	3	3	1	Owner	64	A
	Delivery Size	kg		90,000	90,000	90,000	90,000	6	Calculated		
	Preferred Delivery Method			Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	1	Owner	64	C
	Bulk Density		tm³	0.95	0.95	0.95	0.95	1	Owner	16	A
	Solids SG			3.37	3.37	3.37	3.37	1	Owner	16	A
	Water SG			1.07	1.07	1.07	1.07	1	Owner	16	A
	Solution Mix Strength	% w/w		20	20	20	20	1	Owner	16	A
	Lime Slurry Density	tm³		1.24	1.24	1.24	1.24	6	Calculated		Site metallurgical accounting assumption Cyclone overflow density
	Feed Size										
	F100		mm	6.7	6.7	6.7	6.7	1	Owner	116	1
	F80		µm	1,150	1,150	1,150	1,150	1	Owner	116	1
	Product Size		µm	38	38	38	38	1	Owner	116	1
	Dry Consumption										
	Concentrate CIL - Nominal	concentrate basis	kg/t	51.2	51.2	51.2	51.2	1	Owner	23	A
	Concentrate CIL - Design	concentrate basis	kg/t	120.1	120.1	120.1	120.1	1	Owner	16	A
	Flotation Tailings CIL - Nominal	tailings CIL basis	kg/t	3.20	3.20	3.20	3.20	1	Owner	23	A
	Flotation Tailings CIL - Design	tailings CIL basis	kg/t	4.50	4.50	4.50	4.50	1	Owner	16	A
	Total Dry Consumption										
	Nominal	@ 93.4% Availability	th	13.9	14.8	15.4	15.6	6	Calculated		
	Nominal	@ 75% Lime Utilisation	th	17.3	18.5	19.2	19.3	6	Calculated		75% lime plant utilisation as per 2000-F72-TQY-PR-15033
	Design	@ 93.4% Availability	th	23.2	25.9	26.7	25.8	6	Calculated		
	Design	@ 75% Lime Utilisation	th	28.9	32.2	33.2	32.1	6	Calculated		75% lime plant utilisation as per 2000-F72-TQY-PR-15033
	Milk of Lime Consumption										
	Nominal		m³/h	56.2	59.9	62.2	62.4	6	Calculated		
	Peak		m³/h	93.5	104.4	107.7	104.0	6	Calculated		
	New Circuit										
	Storage Silo Capacity	per Silo	t	1,132	1,132	1,132	1,132	5	Vendor	118	1
	Number of Silos			2	2	2	2	5	Vendor	118	1
	Residence Time	Actual	d	7	6	6	6	6	Calculated		Based on nominal consumption
	Lime Slaker Type			MIII	MIII	MIII	MIII	1	Owner	43	A
	Number of Slakers			2	2	2	2	5	Vendor	118	1
	Slaker Power		kW/mill	6 x 55	6 x 55	6 x 55	6 x 55	5	Vendor	118	2
	Utilisation	%		75	75	75	75	1	Owner	116	A
	Mill Throughput	Design	thr/mill	17	17	17	17	5	Vendor	118	2
			t/hr	33	33	33	33	5	Vendor	118	2
	Make-up Water Type			Raw Water	Raw Water	Raw Water	Raw Water	1	Owner	103	A
	Milk of Lime Storage Tank Live Volume	Max	m³	524	524	524	524	1	Owner	107	0
	Number of Storage Tanks			1	1	1	1	1	Owner		A
	Residence Time	Actual	h	9.3	8.8	8.4	8.4	6	Calculated	43	
	Ringmain Flow Factor			2.5	2.5	2.5	2.5	2	EPC		C
	Lime Ringmain Pumps	Required	m³/h	234	261	269	260	6	Calculated		
		Selected	m³/h	300	300	300	300	2	EPC		C

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
19.12	Hydrochloric Acid			New							
	Delivery Form			Bulk Liquid				1	Owner	64	A
	Delivery Size per Trailer			20,000				1	Owner	64	A
	Number of Trailers per delivery	Maximum		3				1	Owner	64	A
	Delivery Size	L		60,000				6	Calculated		A
	Preferred Delivery System			Pumped Transfer From Truck				1	Owner	64	A
	Delivery Side			Passenger				1	Owner	64	B
	Supply Strength	% w/w		32				1	Owner	64	A
	Supply Solution SG	t/m³		1.16				1	Owner	16	A
	Consumption										
	Acid Wash Column 1	m³/strip		2.7				1	Owner	16	A
		Min. instantaneous		9.5				6	Calculated		
	Acid Wash Column 2	m³/strip		2.7				1	Owner	16	A
		Min. instantaneous		9.5				6	Calculated		
	Acid Wash Column CIL4	m³/strip		1.5				6	Calculated		
		Min. instantaneous		5.2				6	Calculated		
	Consumption										
	Acid Wash Column 1	kg/d		1,503				6	Calculated		
	Acid Wash Column 2	kg/d		1,503				6	Calculated		
	Acid Wash Column CIL4	kg/d		1,085				6	Calculated		
	Calculated Consumption										
		kg/d		4,092				6	Calculated		
		kg/h		170				6	Calculated		
		L/h		256				6	Calculated		
		L/h		459				6	Calculated		
	New Circuit										
	Number of Storage Tanks			1				1	Owner	24	A
	Live Volume per Tank	m³		200				2	EPC		A
	Residence Time	Nominal	d	18				6	Calculated		
										Based on Nominal Consumption	
	Dilute Acid Storage Tank Live Volume	m³		25				2	EPC	110	0
	Neat Acid Dosing Pumps	Required	m³/h	20				2	EPC		C
	Dilute Acid Dosing Pump	Minimum	m³/h	64				2	EPC		0
19.13	Grinding Media										
	SAG Mill										
	Number of Bunkers			2	2	2	2	1	Owner	16	A
	Bunker Storage Size	m³		35	35	35	35	1	Owner	16	A
	Media Type			Forged Steel	Forged Steel	Forged Steel	Forged Steel	1	Owner	16, 38	A
	Media Diameter	mm		125	125	125	125	1	Owner	16, 38	A
	Bulk Density	t/m³		4.65	4.65	4.65	4.65	1	Owner	16	A
	Package Form			Bulk	Bulk	Bulk	Bulk	1	Owner	16	A
	Delivery Size	t		25	25	25	25	1	Owner	23	A
	Trailers per Delivery	#		1	1	1	1	1	Owner	23	A
	Consumption Rate - Existing SAG	kg/t mill feed		0.35	0.35	0.35	0.35	1	Owner	16, 38	A
	Consumption Rate - New SAG	kg/t mill feed		0.40	0.40	0.40	0.40	1	Owner	16, 38	A
	Total Consumption	t/a		9,759	10,649	10,724	10,899	6	Calculated		
	Storage Capacity	t		325.5	325.5	325.5	325.5	6	Calculated		
	Residence Time	d		12	11	11	11	6	Calculated		
	Indexer/Charger Storage Capacity	Minimum	Buckets	1.5	1.5	1.5	1.5	1	Owner	16	CAT972 Bucket
	Ball Mill										
	Number of Bunkers			1	1	1	1	1	Owner	16	A
	Bunker Storage Size	m³		35	35	35	35	1	Owner	16	A
	Media Type			High Chrome	High Chrome	High Chrome	High Chrome	1	Owner	16, 38	A
	Media Diameter	mm		75	75	75	75	1	Owner	16, 38	A
	Bulk Density	t/m³		4.75	4.75	4.75	4.75	1	Owner	16	A
	Package Form			Bulk	Bulk	Bulk	Bulk	1	Owner	16	A
	Delivery Size	t		25	25	25	25	1	Owner	23	A
	Trailers per Delivery	#		1	1	1	1	1	Owner	23	A
	Consumption Rate	kg/t mill feed		0.20	0.20	0.20	0.20	1	Owner	16, 38	A
	Consumption	t/a		5,148	5,699	5,699	5,718	6	Calculated		
	Storage Capacity	t		166.3	166.3	166.3	166.3	6	Calculated		
	Residence Time	d		12	11	11	11	6	Calculated		
	Indexer/Charger Storage Capacity	Minimum	Buckets	1.5	1.5	1.5	1.5	1	Owner	16	CAT972 Bucket
	UFG Mill										
	Media Type			Ceramic	Ceramic	Ceramic	Ceramic	1	Owner	16	A
	Media Diameter	mm		2.5	2.5	2.5	2.5	1	Owner	16	A
	Bulk Density	t/m³		2.2	2.2	2.2	2.2	1	Owner	16	A
	Package Form			Bulk Bag	Bulk Bag	Bulk Bag	Bulk Bag	1	Owner	16	A
	Package Weight	kg		2,000	2,000	2,000	2,000	1	Owner	16	A
	Addition Rate	kg/t mill feed		0.56	0.56	0.56	0.56	1	Owner	10	B
		UFG Mill 1		0.0	6.8	6.8	0.0	6	Calculated		
		UFG Mill 1		0.0	163.0	163.2	0.0	6	Calculated		
		UFG Mill 2		21.0	25.8	25.8	23.4	6	Calculated		
		UFG Mill 2		503.4	618.2	618.2	561.2	6	Calculated		
		UFG Mill 3		21.0	25.8	25.8	23.4	6	Calculated		
		UFG Mill 3		503.4	618.2	618.2	561.2	6	Calculated		
	Site Storage Capacity	Minimum	# Bags	50	50	50	50	1	Owner	10	B
		Design	# Bags	54	54	54	54	2	EPC		
		days		107	77	77	96	6	Calculated		
	Lime Slaking Mills										
	Media Type			Steel	Steel	Steel	Steel	5	Vendor	118	2
	Media Diameter	mm		25.0	25.0	25.0	25.0	5	Vendor	118	2
	Bulk Density	t/m³		4.85	4.85	4.85	4.85	5	Vendor	118	2
	Total Consumption	kg/mill/week		170	170	170	170	5	Vendor	118	2
		kg/week total		340	340	340	340	6	Calculated		
		kg/t		10.2	10.2	10.2	10.2	6	Calculated		
	Package Form			Bulk Bag	Bulk Bag	Bulk Bag	Bulk Bag	1	Owner	118	2
	Media Hopper Capacity	UFG Mill 2	m³	14	14	14	14	5	Vendor	65	B
	Residence Time	UFG Mill 2	days	61.2	49.8	49.8	54.9	6	Calculated		
	Residence Time	UFG Mill 2	# Bags	15.4	15.4	15.4	15.4	6	Calculated		
	Media Hopper Capacity	UFG Mill 3	m³	14	14	14	14	5	Vendor	65	B
	Residence Time	UFG Mill 3	days	61.2	49.8	49.8	54.9	6	Calculated		
	Residence Time	UFG Mill 3	# Bags	15.4	15.4	15.4	15.4	6	Calculated		
19.14	Antiscalant										
	Type			FF585				1	Owner	16	
	Addition Rate - Gravity Concentrators	ml/min		40				1	Owner	16	
	Addition Rate - Kiln Scrubber (Existing)	ml/min		45				1	Owner	16	
	Addition Rate - CIL4 Kiln Mercury Abatement System	ml/min		61				1	Owner	44	B
	Total Consumption	ml/min		146				6	Calculated		
	Number of IBC's for Storage	No.		2				6	Calculated		
	Type			FF540				1	Owner	16	
	Addition Rate - Elution Circuit 1 (Existing)	ml/min		56				1	Owner	16	
	Addition Rate - Elution Circuit 2 (Existing)	ml/min		70				1	Owner	16	
	Addition Rate - Elution Circuit 1 Heat Exchanger (Existing)	ml/min		24				1	Owner	16	
	Addition Rate - Elution Circuit 2 Heat Exchanger (Existing)	ml/min		30				1	Owner	16	
	Addition Rate - Elution Circuit CIL4 (New)	ml/min		95				1	Owner	44	B
	Addition Rate - Elution Circuit CIL4 Heat Exchanger (New)	ml/min		41				1	Owner	44	B
	Total Consumption	ml/min		316				6	Calculated		
	Number of IBC's for Storage	No.		4				6	Calculated		
19.15	LPG										
	Type			Propane				1	Owner	16	A
	Elution Burner 1 Range	High Flame	MJ/h	4,900				1	Owner	16	
		Low Flame	MJ/h	2,000				1	Owner	16	
	Elution Burner 2 Range	High Flame	MJ/h	4,900				1	Owner	16	
		Low Flame	MJ/h	2,000				1	Owner	16	
	CIL4 Elution Burner Range	Pre-Heat	L/h	823				5	Vendor	81	
		Steady State	L/h	708				5	Vendor	81	
	Regeneration Kiln 1		MJ/h	4,500				1	Owner	16	
	Regeneration Kiln 2		MJ/h	4,500				1	Owner	16	
	Regeneration Kiln 3		MJ/h	4,500				1	Owner	16	
	Regenerative Thermal Oxidiser		MJ/h	1,580				1	Owner	88	B
	Smelting Furnace	Maximum	MJ/h	2,000				1	Owner	16	
	Consumption Rate	Plant Feed Basis	L/t	0.40				6	Calculated	50	
										Inclusive of Regeneration Kiln 3 and RTO but not the Goldroom Smelting Furnace. Calculated from a 25.5 Mtpa basis.	B
	Number of LPG Storage Vessels (Upgraded)		kL	1				1	Owner	88	B
	Vessel Capacity			161				1	Owner	88	B
	Residence Time	Actual	d	5.3				6	Calculated		
										As per source 82, Owner confirm and accept they won't achieve the 7 days	

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIM	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
20.0	WATER SERVICES - EXISTING										
20.1	Saline Bore Water										
	Source			Borefields			1	Owner	16		A
	Storage Method 1			Dam 1 (North)			1	Owner	16		A
	Live Volume	m³		8,000			1	Owner	16		A
	Dam Water Pumps	#		2			1	Owner	23		A
	Pump Model			Stalker 250-250-400, 132kW			1	Owner	23		A
	Storage Method 2			Dam 2 (South)			1	Owner	16		A
	Live Volume	m³		12,000			1	Owner	16		A
	Dam Water Pumps	#		3			1	Owner	106		O
	Pump Model			TKL 250x250 - 400HT			1	Owner	23		A
	Total Live Volume	m³		20,000			6	Calculated			
20.2	Mill Circuit Water										
	Source			Concentrate Pre-Leach Thickener O/F Concentrate Thickener O/F	Concentrate Pre-Leach Thickener O/F Concentrate Thickener O/F	Concentrate Pre-Leach Thickener O/F Concentrate Thickener O/F	1	Owner			B
	Storage Method			Circuit Water Tank	Circuit Water Tank	Circuit Water Tank	1	Owner	16		A
	Live Volume	m³		1,500	1,500	1,500	1	Owner	16		A
	Residence Time	min		100	94	94	6	Calculated			
	Water Demand - Nominal	Calculated	m³/h	896	958	958	6	Calculated			
	Water Demand - Design		m³/h	1,800	1,800	1,800	1	Owner	16		A
	Design Flowrate per Pump		m³/h	1,800	1,800	1,800	1	Owner	16		A
	Pump Configuration			Duty / Standby	Duty / Standby	Duty / Standby	1	Owner	16		A
	Pump Model			APP51-300	APP51-300	APP51-300	1	Owner	23		A
20.3	Cyanide Water										
	Source			Decant Return			1	Owner	16		A
	Storage Method			Cyanide Water Tank			1	Owner	16		A
	Live Volume	m³		1,115			1	Owner	16		A
	Cyanide Water Pump Configuration			Duty / Standby			1	Owner	16		A
	Cyanide Water Pump Model			TKL 200x200 - 500HT			1	Owner	23		A
	Cyanide Transfer Pump Configuration			Duty / Standby			1	Owner	16		A
	Cyanide Transfer Pump Model			TKL 100x65 - 315HT			1	Owner	23		A
20.4	Scheme Water										
	Water Demand		m³/h	172	172	173	6	Calculated			
	Scheme Water Quality										
	SG	-		1.00	1.00	1.00	1	Owner	1		B
	TDS	mg/L		454	454	454	1	Owner	1		B
	pH	-		8.0	8.0	8.0	1	Owner	1		B
	Chlorides	mg/L	Range	232	232	232	1	Owner	1		B
20.5	Fire Water / Safety Shower / Potable Water						1	Owner	69	By Others	B
20.6	Gland Water										
	Source			Dam 2 (South)			1	Owner	16		A
	Storage Method			Gland Water Tank			1	Owner	16		A
	Live Volume	m³		83			1	Owner	16		A
	Pump Configuration			Duty / Standby			1	Owner	16		A
	Pump Model			TIK 90kW			1	Owner	23		A
20.7	Eastern Borefield Transfer Dam										
	Source			Seepage			1	Owner	16		A
	Storage Method			Dam			1	Owner	16		A
	Destination			Dam 3			1	Owner	79		B
	Live Volume	m³		12,000			1	Owner	16		A
	Pump Configuration			Duty / Standby			1	Owner	16		A
20.8	Saline Gravity Dam										
	Source			Borefield			1	Owner	16		A
	Storage Method			Dam			1	Owner	16		A
	Destination			Dam 1 (North)			1	Owner	79		B
	Live Volume	m³		16,000			1	Owner	16		A
20.9	Kahtais Saline Dam										
	Source			Borefield			1	Owner	16		A
	Storage Method			Dam			1	Owner	16		A
	Destination			Dam 3			1	Owner	79		B
	Live Volume	m³		13,600			1	Owner	16		A
	Pump Configuration			Duty / Duty / Standby			1	Owner	16		A
21.0	Saline Transfer Dam										
	Source			Borefield			1	Owner	16		A
	Storage Method			Dam			1	Owner	16		A
	Destination			Dam 1 (North) and Dam 2 (South)			1	Owner	79		B
	Live Volume	m³		340			1	Owner	16		A
	Pump Configuration			Duty / Standby			1	Owner	16		A
20.10	Southern Surge Tank										
	Storage Method			Southern Surge Tank			1	Owner	16		A
	Destination			Dam 2 (South)			1	Owner	16		A
	Capacity	m³		675			1	Owner	16		A
21.0	WATER SERVICES - NEW										
21.1	Process Water										
	Source			Flotation Tailings Thickener O/F Saline Bore Water Makeup	Flotation Tailings Thickener O/F Saline Bore Water Makeup	Flotation Tailings Thickener O/F Saline Bore Water Makeup	1	Owner	10		A
	Storage Method			Tank	Tank	Tank	9	Agreed	10		A
	Live Volume	m³		2,080	2,080	2,080	9	Agreed	10		A
	Residence Time	Minimum	min	15	15	15	1	Owner	16		A
	Water Demand	Calculated	m³/h	5,355	5,152	5,489	6	Calculated			
	Pump Configuration			Duty / Standby	Duty / Standby	Duty / Standby	9	Agreed	10		A
	Process Water Quality										
	SG	-		1.07	1.07	1.07	1	Owner	1		B
	TDS	mg/L		140,000	140,000	140,000	1	Owner	1		B
	pH	-		4.6	4.6	4.6	1	Owner	1		B
	Chlorides	mg/L	Range	20,492 - 86,243	20,492 - 86,243	20,492 - 86,243	1	Owner	1		B
21.2	Raw Water										
	Source			Dam 2 (South)	Dam 2 (South)	Dam 2 (South)	1	Owner	43	Supplies the Raw Water Tank	A
	Storage Method			Steel Tank	Steel Tank	Steel Tank	1	Owner	68		B
	Configuration			Separate Raw, UFG 2/3 Gland, LP and HP Gland	Separate Raw, UFG 2/3 Gland, LP and HP Gland	Separate Raw, UFG 2/3 Gland, LP and HP Gland	1	Owner	68		B
	Total Live Volume	Actual	m³	375	375	375	1	Owner	68		B
	Total Live Volume	Required	m³	377	377	377	6	Calculated			
	Residence Time	Actual	min	60	60	60	6	Calculated			
	Residence Time	Required	min	60	60	60	9	Agreed	10, 16	Nominal Basis	B
	Total Nominal Water Demand (water consumption)		m³/h	377	377	377	2	EPC	68	Note: Pump design for instantaneous flows in accordance with mechanical calculation.	B
	Pump Configuration			Duty / Standby	Duty / Standby	Duty / Standby	1	Owner	16		A
	Source (Greenfields)			Dam 3	Dam 3	Dam 3	1	Owner	69	By Others	B
	Live Volume	m³		35,000	35,000	35,000	1	Owner	114	By Others	1
21.3	Gland Water										
	Source			Raw Water Tank	Raw Water Tank	Raw Water Tank	1	Owner	43		A
	Gland Water Demand	Design	m³/h	92	92	92	2	EPC			O
	Low Pressure Gland Water Pump Configuration			Duty / Standby	Duty / Standby	Duty / Standby	1	Owner	43		A
	High Pressure Gland Water Pump Configuration			Duty / Standby	Duty / Standby	Duty / Standby	1	Owner	43		A
22.0	AIR SERVICES										
22.1	Low Pressure Flotation Air										
	Number of Flotation Blowers			4			1	Owner	49		A
	Operating Configuration			Duty / Duty / Duty / Standby			1	Owner	49		A
	Type			Low pressure, oil free screw blower with sound enclosure			1	Owner	49		A
	Capacity - Each	Design	FAD m³/h	8,688			1	Owner	49		A
	Capacity - Total		FAD m³/h	26,064			6	Calculated			
	Pressure	Design	kPa(g)	119.7			1	Owner	49		A
22.2	High Pressure Air										
	Number of Compressors			3			1	Owner	43		A
	Operating Configuration			Duty / Duty / Standby			1	Owner	43		A
	Type			Oil Injected Rotary Screw Compressor			5	Vendor	45		B
	Capacity - Each	NTP Conditions	L/s	750			5	Vendor	45		A
	Pressure		kPa(g)	980			5	Vendor	45		B
	Duty			New facilities only			2	EPC			A